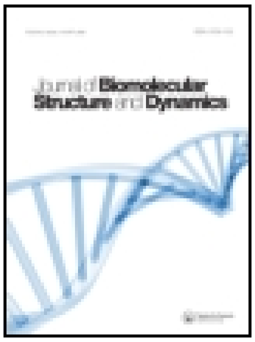

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Screening of Promising Molecules against MurG as Drug Target in Multi-Drug-Resistant-*Acinetobacter baumannii* - Insights from Comparative Protein Modeling, Molecular Docking and Molecular Dynamics Simulation

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Abstract

The UDP-*N*-acetylglucosamine-*N*-acetylmuramyl-(pentapeptide) pyrophosphoryl-undecaprenol *N*-acetylglucosamine transferase (MurG) is located in plasma membrane which plays a crucial role for peptidoglycan biosynthesis in Gram-negative bacteria. Recently, this protein is considered as an important and unique drug target in *Acinetobacter baumannii* since it plays a key role during the synthesis of peptidoglycan as well as which is not found in *Homo sapiens.* In this study initially we performed comparative protein modeling approach to predict the three-dimensional model of MurG based on crystal structure of UDP-*N*-acetylglucosamine-*N*-acetylmuramyl-(pentapeptide) pyrophosphoryl-undecaprenol N-acetylglucosamine transferase (PDB ID: 1F0K) from *E.coli* K12. MurG model has two important functional domains located in *N* and *C*- terminus which are separated by a deep cleft. Active site residues are located between two domains and they are Gly20, Arg170, Gly200, Ser201, Gln227, Phe254, Leu275, Thr276, and Glu279 which play essential role for the function of MurG. In order to inhibit the function of MurG, we employed the High Throughput Virtual Screening (HTVS) and docking techniques to identify the promising molecules which will further subjected into screening for computing their drug like and pharmacokinetic properties. From the HTVS, we identified 5279 molecules, among these, 12 were passed the drug-like and pharmacokinetic screening analysis. Based on the interaction analysis in terms of binding affinity, inhibition constant and intermolecular interactions, we selected four molecules for further MD simulation to understand the structural stability of protein-ligand complexes. All the analysis of MD simulation suggested that ZINC09186673 and ZINC09956120 are identified as most promising putative inhibitors for MurG protein in *A. baumanii*.

Keywords: Homology modeling, Hit selection, Virtual screening, *Acinetobacter baumannii*, MD simulation, Molecular Docking, MurG protein.

Abbreviations: MD, Molecular Dynamics; PCA, Principal Component Analysis; ED, Essential Dynamics, MurG, UDP-N-acetylglucosamine-N-acetylmuramyl-(pentapeptide) pyrophosphoryl-undecaprenol N-acetylglucosamine transferase; HTVS, High Throughput Virtual Screening; RMSD, Root Mean Square Deviation; RMSF, Root Mean Square Fluctuation; RoG, Radius of Gyration; SASA, Solvent Accessible Surface Area; PSA, Polar Surface Area

1. Introduction

Acinetobacter baumannii (A. baumannii) is aerophilic, have a thin cell wall and immotile bacteria detected mostly in hospital patients. In the recent healthcare system, this pathogenic bacteria is mainly responsible for hospital acquired nosocomial infections. Due to the characteristics of multi-drug resistance (MDR) of *A. baumannii*, only few antibiotics are available for treating infections caused by this pathogen. This pathogenic bacteria showing the severe resistance against various classes of antibiotics (C.-R. Lee et al., 2017). *A. baumannii* is belongs to ESKAPE (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *A. baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* species) microorganism normally implicated inwards nosocomial contagions. Due to its innate resilience against environmental stresses such as desiccation, nutrient starvation, and acquired resistance to a wide array of antibiotics, including penicillin and cephalosporin, and last resort antibiotics such as polymyxins and carbapenems, *A. baumannii* has been emerged as one of the most successful and challenging pathogens among all groups (Bravo et al., 2016; Haeili, Kafshdouz, & Feizabadi, 2018; C. R. Lee et al., 2017). *A. baumannii* is frequently implicated in pneumonia, bacteraemia, septicaemia, urinary tract infection and meningitis. According to WHO, this bacteria is listed as Priority I and critical pathogen because of the MDR against various classes of antibiotics (Shrivastava, Shrivastava, & Ramasamy, 2018). *A. baumannii is* estimated to be responsible for 2 to 10% of gram-negative infections acquired in Europe and the USA. However, there is growing evidence for increase infection and mortality rates associated with multi-drug resistance *A. baumannii* (MDRAB) in all over the world (Nelson et al., 2016; Teerawattanapong et al., 2018). Therefore it is very important and necessary to identify the new classes of potential drug molecules which treat the infections caused by *A. baumannii.*

Recently few computational studies identified Mur members (MurA and MurG) as potential drug targets towards MDRAB and *Mycobacterium tuberculosis*, respectively (Saxena, Abdullah, Sriram, & Guruprasad, 2018; Skariyachan, Manjunath, & Bachappanavar, 2019). They utilized Structure Based Virtual Screening approaches to identify the promising lead molecules against MurA protein from MDRAB which provided the key structural insights for understanding the efficacy, binding affinity and inhibitory potential of promising molecules towards MurA protein (Skariyachan et al., 2019). Moreover another *in silico* studies identified diverse classes of lead molecules from commercial Asinex database which significantly binds to the active site of MurG protein from *M. tuberculosis* (Saxena et al., 2018). Furthermore other studies also utilized *in silico* approaches and identified some of the effective molecules against efflux pump extensively found in MDRAB (Verma, Maurya, Tiwari, & Tiwari, 2019; Verma, Tiwari, & Tiwari, 2018). Aforementioned reports suggested that *A. baumannii* is most severe pathogen which shows resistance against diverse classes of antibiotics. Among the various pathway found in MDRAB, the peptidoglycan biosynthetic pathway is most important for the survival of the bacteria. Enzymes involved in this bio synthetic pathway is also crucial drug target for the identification of lead molecules which disturb their normal function.

Bacterial peptidoglycan is made up of biological macromolecule that covers cell membranes and components of the cell walls. This bacterial cell wall is responsible for the shapes and firmness as well as protecting cells from burst because of variations in osmotic pressure. In general, murein or peptidoglycan is the chief constituent of the outer membrane of the majority of eubacteria cell walls, including *A. baumannii* (Saxena et al., 2018). Subsequently, the biosynthesis of peptidoglycan is exceptional as well as its role is distinctive to microbial cell component, the proteins that catalyze its production are a striking mark for the design as well as finding of antibiotics for *A. baumannii*. Since the biological enzymes that catalyze the synthesis of murein in *A. baumannii* are of a multiple significance, consequently hang-up of such enzyme can hamper the existence of the *A. baumannii*. Enzymes involved in peptidoglycan biosynthesis are very essential for several bacterial species (Z. Fakhar et al., 2016; Saxena et al., 2018). Among the enzymes involved in this pathway, the transfer of a GlcNAc subunit on undecaprenyl-pyrophosphoryl-MurNAc-pentapeptide (lipid intermediate I) to form undecaprenyl-pyrophosphoryl-MurNAc-(pentapeptide) GlcNAc (lipid intermediate II) is catalyzed by MurG protein. It is transported through the plasma membrane where it is polymerized and join together. It is also the final step of the biosynthesis of the peptidoglycan pathway (Zeynab Fakhar et al., 2016). In addition to this, the protein plays a role of cell growth and survival. This function is conserved among eubacteria including *E. coli*. The activity of this enzyme in *E. coli* is well organized including the overall structure and catalytic site. Consequently, the 3D structure of *E. coli* MurG contains two domains. Domain I is separated by a deep cleft from domain II. The α-helix and β-plated sheet structure is found in both domains. MurG from *A. baumanii* have high structural similarity with MurG from other bacterial species including *E. coli*. In *E. coli*, the nitrogen terminal portion consist of residues from 70 up to 163 and 341 up to 357. Additionally, this domain comprises of seven parallel β-strands and six α-helices. The carboxylic terminal portion of the structure comprises residues from 164 up to 340. This domain also comprises of six parallel β-strands and eight α-helices. Apart from this, in *E. coli* MurG, the critical amino acids involved in the UDP-GlcNAc catalytic site include Ala264, Asn292, Gly263, Gln289, Ala293, Glu269, Arg261, Pro281, and Asn199. In addition to this, the residues involved in the di-phosphate binding site are Gly102, His125, Glu126, Asn128, Gly132, and Asn135 (Ha, Walker, Shi, & Walker, 2000). The MurG protein from MDRAB has served as the unique potential target for drug designing and discovery process, since MurG plays a key functional role in the synthesis of the peptidoglycan in *A. baumannii* as well as it does not found in animal cells. Based on above importance of MurG with reference to structure and function, in the present study we chosen this as an important drug target for *A. baumannii*.

The present study explored the sequential and structural properties of MurG drug target protein from MDRAB and thus allowed us to facilitate the computer aided virtual screening to identify novel drug like molecules against this protein by comparative protein modelling, binding site analysis, molecular docking and molecular dynamics simulations. Together the results obtained from all these analysis, we identified some promising molecules which could serve as potential leads for the discovery and development of inhibitors against MDRAB-MurG useful for the treatment of hospital acquired nosocomial infections.

2. Materials and methods

2.1. Protein sequence analysis

The amino acid sequence of MurG from *A. baumannii* was retrieved from UniProt (https://www.uniprot.org) with the accession number of B0V9F5. Subsequently, the protein sequences were subjected in to protein sequence analysis using Expert Protein Analysis System (ExPASy) Proteomics web server (www.expasy.org/tools) (Gasteiger et al., 2005). The physico-chemical properties such as amino acid composition, molecular weight (MW), isoelectric point (pI), instability index (II), aliphatic index (AI) and extinction coefficient (EC) was computed using ProtParam web server (Gasteiger et al., 2005). In addition, we used Simple Modular Architecture Research Tool (SMART) to identify the functional domains present in the protein sequence (Letunic, Doerks, & Bork, 2015). PSort server (http://psort.hgc.jp/) was also used to predict the sub-cellular localization of the protein (Yu et al., 2010). Moreover, the EMBOSS-

Antigenic program (Rice, Longden, & Bleasby, 2000) was employed to predict the possible antigenic sites of MurG protein. Antigenic sites are very important region which determining the functionality of the bacterial enzymes.

2.2. Homology Modeling

For template selection, Protein BLAST (Basic Local Alignment Search Tool for protein) (https://blast.ncbi.nlm.nih.gov/Blast.cgi) (Altschul et al., 1997) was used to identify the homologous sequences of MurG with the aid of Protein Database (PDB) (https://www.rcsb.org/structure/) (Berman et al., 2000) on the basis of sequence identity, similarity and query coverage. Clustal Omega (https://www.ebi.ac.uk/Tools/msa/clustalo/) was used to perform the multiple sequence alignment of target and its homologous sequences which are obtained from Protein BLAST search (Madeira et al., 2019). Followed by, an un-rooted phylogenetic tree was constructed for MurG and its related four homologs by Molecular Evolutionary Genetic Analysis Version X (MEGA X) program using four methods namely Neighbor Joining (NJ), Maximum Parsimony (MP), Minimum Evolution (ME) and Maximum likelihood (ML) (S. Kumar et al., 2018). The best template was selected and validated on the basis of following parameters namely: (a) higher resolution of template structure, (b) better query coverage between target and template sequence, (c) sequence identify of above 40% between template and target sequence and (d) both target and template found in same clade or same group of cladogram.

The three-dimensional structure of MurG was modeled on the basis of the crystal structure of UDP-N-acetylglucosamine-n-acetylmuramyl-(pentapeptide) pyrophosphoryl-undecaprenol n-acetylglucosamine transferase (PDB ID: 1F0K) from *Escherichia coli* (Strain K12) obtained from Protein BLAST result, thus used as template for the modeling study. The three-dimensional model of MurG was predicted by modelling package Modeller 9.21 (Webb & Sali, 2016) based on satisfaction of spatial restraints. The Align script (align.py) of Modeller program was used to align the MurG sequence with template structure. The sequence identity of alignment between the query sequence and the template determines the quality of the model structure performed so that this step is a prerequisite to the success of this technique. Based on the sequence alignment information and template structure, the model-single python script (model-single.py) (Webb & Sali, 2016) was used to build the five different three-dimensional models of MurG. The best model was selected based on three parameters such as modeller objective function, GA341 and Discrete Optimized Protein Energy (DOPE). Subsequently the following four programs, namely: YASARA Energy Minimization Server (Krieger et al., 2009), Chiron protein structure refinement server (Ramachandran, Kota, Ding, & Dokholyan, 2011), modrefiner (high-resolution protein structure refinement) (D. Xu & Zhang, 2011), and 3Drefine (protein structure refinement server) (Bhattacharya & Cheng, 2013; Bhattacharya, Nowotny, Cao, & Cheng, 2016) were used to obtain the optimal and reliable model for further *in silico* interaction studies with various ligands. For understanding the structural similarity between template and predicted model, we used both FATCAT (Li, Ye, & Godzik, 2006; Ye & Godzik, 2004) and PyMOL programs (DeLano, 2002). Earlier one was utilized to perform the pairwise structure alignment between template and predicted model of MurG and later one was used to perform superposition between two structures to compute the RMSD (Root Mean Square Deviation) values.

2.3. Model evaluation and putative binding site prediction

The optimized 3D model obtained from the previous step was subjected into various servers for model validation such as PROCHECK (R. A. Laskowski, MacArthur, Moss, & Thornton, 1993; Roman A Laskowski, Rullmann, MacArthur, Kaptein, & Thornton, 1996), ERRAT (Colovos & Yeates, 1993) and Verify_3D (Bowie, Luthy, & Eisenberg, 1991; Luthy, Bowie, & Eisenberg, 1992). The Procheck (R. A. Laskowski et al., 1993; Roman A Laskowski et al., 1996) was used to determine the overall structural quality assessment (stereochemistry) based on different type of internal angles, main chain ionic forces and interatomic interaction distances through the Ramachandran plot. The ERRAT program (Colovos & Yeates, 1993) was used to distinguish and analyzes correctly and inappropriately resolute segments of a modeled configurations on the basis of their characteristic atomic relations using statistical approaches. Finally, the Verify_3D (Bowie et al., 1991; Luthy et al., 1992) was used to evaluate the compatibility among 3-D residues with 1-D residues, which is shown by secondary structure (α-spirals, β-sheets, loop), soluble and insoluble amino acid using water, as well as at the end connecting the outputs with proper configurations.

The possible binding sites of MurG was predicted using CastP (Tian, Chen, Lei, Zhao, & Liang, 2018), PockDrug (Hussein et al., 2015), 3DLigandSite (Wass, Kelley, & Sternberg, 2010), and CavityPlus (Y. Xu et al., 2018). Common binding site residues obtained from all these prediction programs are considered as a possible binding site and thus will further validate by performing the pairwise sequence alignment between target and template sequences which active site was already known.

2.4. High-throughput Virtual screening (HTVS) and molecular docking

The library of natural compounds which contain 30,792 ligands was obtained from a special subset of a ZINC database with SDF file format (J. J. Irwin & Shoichet, 2005; John J. Irwin, Sterling, Mysinger, Bolstad, & Coleman, 2012). On the other hand the standard antibiotics which inhibit the growth of MDRAB were used as control for this study namely clinafloxacin (PubChem: CID: 60063) and imipenem (PubChem: CID: 104838) (Skariyachan et al., 2019). HTVS was performed to identify the potential lead compounds with the strongest binding affinity, higher number of intermolecular interactions and significant drug-like properties. The natural compounds of the ZINC database and standard two antibiotics were screened against a MurG model to find lead molecules. In order to improve the average precision of the binding pose predictions, PyRx tool (Dallakyan & Olson, 2015) was used with PDBQT (XYZ coordinates + Charge + Atom type) file format of all natural compounds and MurG model to perform structure-based virtual screening. In this virtual screening, Auto Dock Vina was used to perform the direct docking calculation based on binding site of MurG for all the ligands as well as two antibiotics. The molecular docking calculation is widely used to understand the binding orientation, mode of binding and inter-molecular interactions of one molecule bound to another, in this regard mostly the large molecule as proteins and small molecules as natural compounds. In this calculation, the grid box was created based on binding site of MurG with the following parameters with grid centered on the coordinates of X = -3.64 Å, Y = 2.13 Å and Z = 3.62 Å and grid box size having dimension of X = 25.00 Å, Y= 29.03 Å, Z= 25.00 Å. Based on the values of estimated free energy of binding and estimated inhibition constant, the top eighty six (86) hits were selected for further studies. The physicochemical as well as drug-likeliness properties of the natural compounds were filtered using DataWarrior (http://www.openmolecules.org/datawarrior/) (Veber et al., 2002) as well as PreADMET (Lee et al., 2004) programs.

2.5. Pharmacokinetic analysis

Mostly numerous ligands are undesirable for ADME (absorption, distribution, metabolism, excretion) and toxicity tests which hamper them from the various phases of clinical trial. So that, identification of novel ligands from computational means was used to understanding the pharmacokinetic properties. The ADME and harmful properties of the high binding affinity ligands were checked through the AdmetSAR tool (http://lmmd.ecust.edu.cn/admetsar2) (Cheng et al., 2012) and the DataWarrior tool (http://www.openmolecules.org/datawarrior/) (Veber et al., 2002). The following properties were considered for *in silico* pharmacokinetics screening using AdmetSAR tool, namely Blood-Brain Barrier penetration (BBBP), Human Intestinal Absorption (HIA), Renal Organic Cation Transporter (ROCT), Cytochrome P450 (CYP450 2D6) Inhibitor, AMES Toxic, Carcinogens and Aqueous Solubility. In addition to this, DataWarrior tool was used to study Mutagenicity, Tumorigenicity, Carcinogenicity, Reproducibility, as well as Irritation properties of the ligands. Together with the results obtained from aforementioned screening, the best molecules will be subjected for MD simulation to understand the structural stability, integrity, flexibility and compactness of protein-ligand complex.

2.6. Analysis of HTVS and docking results

PyMOL Molecular Graphics System, Version 2.3. Schrodinger, LLC (DeLano, 2002), Discovery Studio (DS) visualizer (BIOvIA, 2015) and Ligplot$^+$ v.2.1 tool (R. A. Laskowski & Swindells, 2011; Wallace, Laskowski, & Thornton, 1995) were used for the visualization of docking orientation of ligands with the binding site of MurG. In addition to visualization, these software used to measure hydrogen bonds and hydrophobic interactions between protein and ligand molecule. The schematic representation of protein-ligand complex was prepared by discovery studio and Ligplot$^+$ v.2.1.

2.7. Molecular dynamic (MD) simulation

The AmberTools18 package was used to perform the Molecular Dynamics Simulation to investigate the structural stability of MurG and MurG-Ligand complexes (Case et al., 2015) at 300 K. The addition of explicit hydrogen to the MurG modeled structure and MurG–ligand complex through protonate 3D were the initial step to begin this procedure. Antechamber was used to generate all the missing features of natural compounds. The topology as well as coordinate files of proteins and drug molecules in Amber were constructed using the ff14SB and the GAFF force field, respectively. In order to mimic the biological function of MurG and inhibitory effects of natural compounds at the physiological condition, we performed the MD simulation in aqueous environment. The whole system was introduced into a water environment with the addition of Cl^- ions in the octahedral box of 12 Å with water type of TIP3PBOX. In order to get appropriate geometry and avoid the steric clashes, we subjected the entire system into energy minimization followed by equilibration. This process is completed by minimization of the system with an alternative 5000 steps of conjugate gradient and steepest descent as well as with a restraint run at 544 kcal/mol/Å. Additionally, the system minimized for 2500 steps of steepest descent and conjugate gradients after elimination of the restraints. The whole system was heated using a Langevin dynamic temperature regulation through the initial and final temperature of 0.0 k and 300 k, respectively for 100,000 steps. The collision frequency of Langevin thermostat was fixed at 1 ps without pressure regulator. At fixed conditions of temperature and pressure of 300 K temperature and 1 atm pressure, the production of MD simulation was investigated, using a time step of 2 fs. Additionally, together with the later the Berendsen barostat for steady pressure simulation was used to run the production. The structural stability and integrity of MurG and MurG-ligand complexes was analyzed by Root Mean Square Deviation (RMSD), Root Mean Square Fluctuations (RMSF), Radius of Gyration (RoG) and intermolecular hydrogen bonds. Moreover, we also performed essential dynamics using Principal Component Analysis (PCA) to capture the essential and biological relevant motions of protein and protein-ligand complexes, in this way, we can understand whether ligand enhances the structural stability or not. Finally we also studied the solvent accessible property of protein and protein-ligand complexes. MD Simulation results were analyzed using three program namely VMD, PyMOL and CPPTRAJ.

3. Results and discussion

3.1. Primary sequence analysis MurG protein

The amino acid sequence of MurG from *A. baumannii* was retrieved from UniProt database (https://www.uniprot.org) with the accession number of B0V9F5. The computed molecular weight, amino acid length and theoretical isoelectric point (pI) of MurG protein were 39.35 KDa, 365 and 9.02, respectively. The protein had an instability index of 39.12 and aliphatic index of 97.34. This result revealed that MurG protein is thermodynamically stable.

From the results of motif search and cell localization prediction, this protein has two domains (Glyco-transf-*N* domain and Glyco-transf-*C* domain) and it is a cytoplasmic membrane protein. Moreover, the antigenic sites of MurG protein sequence was also predicted using EMBOSS-Antigenic program (Rice et al., 2000). On the basis of this result, 14 possible antigenic sites were predicted and the results were given in Supplementary section-Table 1. This result revealed that, most of the antigenic sites were found in the biding site of MurG structure which revealed that the residues found in antigenic sites involved interactions with ligand molecules during molecular docking calculation. This showed that antigenic site residues are mostly found in binding site of MurG which are essential for the interactions. In addition to this, the possible binding sites of MurG were predicted and the common binding site residues were obtained from each predicting program results as indicated in the Supplementary section-Table 2. These common residues are considered as a most possible binding site residues and validated by performing the pairwise sequence alignment between MurG and template sequences which active site was already known from literature (Ha et al., 2000). The result showed that most of the residues obtained from sequence alignment is the same with the predicted residues. Overall, Gly20, Arg170, Gly200, Ser201, Gln227, Phe254, Leu275, Thr276, and Glu279 are the common binding site residues found in MurG model.

3.2. Homology modeling, energy minimization and evaluation

The three-dimensional structure of UDP-N-acetylglucosamine--N-acetylmuramyl-(pentapeptide) pyrophosphoryl-undecaprenol N-acetylglucosamine transferase (MurG) is not available in Protein Data Bank (www.rcsb.org). Predicting the structure of pathogenic protein is very important to explore the function, and it allows us to identify the potential ligand molecules using Structure Based Drug Design (SBDD) approach. Hence, in the present study the three-dimensional structure of MurG was predicted based on homology modeling approach with the aid of related structural neighbors or templates available in Protein Data Bank (PDB). Based on the results obtained from Protein BLAST search, we identified four templates which show significant sequence similarity as well as query coverage, namely 3S2U (50.28%, 95%, and 6e-113), 1F0K (43.82%, 97%, and 1e-89), 2IYF (32.56%, 11%, and 0.48) and 4M83 (32.56%, 11%, and 0.49). The suitable structural template was identified based on multiple sequence alignment (Supplementary section-Figure 1) and Phylogenetic analysis using Molecular Evolutionary Genetic Analysis Version X (MEGA X) program with four different methods namely Neighbor

Joining (NJ), Maximum Parsimony (MP), Minimum Evolution (ME) and Maximum likelihood (ML) (Figure 1). Together the results of four phylogenetic methods, we found that MurG is located in the same group of 1F0K and 3S2U. Though, 3S2U is showing higher sequence identity towards target MurG protein, another 1F0K showing higher significant query coverage and good sequence identity. In order to get the more reliable model of MurG, it is always to select higher resolution template structure, in this case the 1F0K structure showing a higher resolution of 1.9 Å (Supplementary section-Figure 2). The output of pairwise sequence alignment indicated that MurG and 1F0K sequences had shares 44.3% (159/359 aa) identity, 61.0% (219/359 aa) sequence similarity, 4.2% (15/359 aa) gaps, 0.5 extended penalties, 97% of query coverage and E-value of 1e-89 (Supplementary section-Figure 3). The overall outcomes of the sequence alignment, BLASTp score, resolution and phylogenetic tree indicated that 1F0K was a suitable structural template for homology modeling.

The protein sequence alignment between MurG and its template (PDB ID: 1F0K) explained that MurG sharing several conserved residues towards template. The conserved sequences include UDP GlcNAc or substrate binding site and the acceptor binding site which is found in the *C*-terminal and *N*-terminal domains respectively (Supplementary section-Figure 3). Based on target-template alignment and 3D structure of the template, five structural models were built and generated using Modeller 9.21 (Webb & Sali, 2016). The regions such as structurally conserved regions (SCRs), structural variable regions (SVRs), R-group, Backbone, *N*- terminal and *C*-terminal as well as angular features were reassigned to the MurG via spatial restraints. The model structure that has least DOPE and GA341 was selected as the best model among them. YASARA (Krieger et al., 2009), Chiron (Ramachandran et al., 2011), Modrefiner (D. Xu & Zhang, 2011) and 3Drefine (Bhattacharya & Cheng, 2013; Bhattacharya et al., 2016) Energy Minimization were performed to minimize the best model structure of MurG before and after model evaluation (Table 1). The alignment of the minimized and evaluated structure of MurG with the template gives raises to two domains, separated by a deep cleft, namely *N*-terminal (domain I) and *C*-terminal (domain II) domains (Figure 2a). The *N*-terminus extends from 1-163 and 353-365 residues, whereas the *C*-terminus consists of residues from 164-352. In addition to this, critical residues that have an important role in the UDP-GlcNAc binding site were Gly20, Arg170, Gly200, Ser201, Gln227, Phe254, Leu275, Thr276, and Glu279 (Figure 2c). Moreover, the predicted model of MurG belongs to the alpha and beta (α/β) class which is showed in the

Supplementary section-Figure 4. For understanding reliability of predicted model of MurG, we performed pairwise structure alignment as well as structure supposition analysis between template and predicted model. The results of pairwise structure alignment explained that both MurG and 1F0K structures are significantly similar with P-value of 0.00e+00 (Supplementary section-Figure 7) and the superimposition of MurG with 1F0K gave a root mean square deviation (RMSD) value of 0.475 Å, this indicated that the predicted model of MurG was reliable and similar to the template (PDB ID: 1F0K) (Figure 2b). Existing reports revealed that, the 3D structure of MurG from *E. coli* contain two domains unglued using a deep cleft. The domain I and domain II consist of α-helices and β-pleated sheet structure and exhibited good structural similarity with MurG from other bacterial species. The residues 7–163 and 341–357 including seven parallel β-pleated sheet and six α-helices belong to the N-terminal domain, whereas residues 164–340 including six parallel β-pleated sheet and eight α-helices are comprised under C-terminal domain. The prominent residues participated in the UDP-GlcNAc binding site are Ala264, Asn292, Gly263, Gln289, Ala293, Glu269, Arg261, Pro281, and Asn199. On the other hand the residues participated in the di-phosphate binding site are Gly102, His125, Glu126, Asn128, Gly132, and Asn135 (Ha et al., 2000).

The overall structural quality of the modeled structure was performed using PROCHECK (Ramachandran plot statistical parameters, Φ-ψ plot and G-factor) (R. A. Laskowski et al., 1993; Roman A Laskowski et al., 1996), Verify_3D (Bowie et al., 1991) as well as ERRAT (Colovos & Yeates, 1993) assessing servers in two phases namely (a) before minimization and (b) after minimization (Supplementary section-Figure 5). Prior to energy optimization the results of the stereochemistry assessment of the MurG model showed that two hundred eighty eight residues (90.9%) were falling in the most favorable region, twenty six residues (8.2%) were in additionally allowed region, two residues (0.6%) were in the generously allowed region, and one residue (0.3%) was in the disallowed region. The minimization of modeled structure was carried out using the following web server namely, YASARA (Krieger et al., 2009), Chiron (Ramachandran et al., 2011), modrefiner (D. Xu & Zhang, 2011) and 3Drefine (Bhattacharya & Cheng, 2013; Bhattacharya et al., 2016). From the results of YASARA server, on the basis of Ramachandran plot analysis, the optimized modeled of MurG and template had 93.7% and 94.6%; 5.4% and 5.4%; 0.6% and 0.0%, and 0.3% and 0.0% of all their amino acid residues were under the most favorable regions, in the extra permitted region, generously permitted region and rejected areas respectively (Table 1). On the basis of Ramachandran plot analysis, more than or equals to 90% of all the residues found in the most favored regions which indicated that the predicted model is considered as a good quality and reliable. In addition, the overall distribution of the amino acid residues of the YASARA optimized MurG model is an indicator of the quality of the model with high structural integrity and compactness. From the Ramachandran plot analysis, the unusual property of the modeled structure was measured using G-factor. If the G-factor values below -0.5, the structure is as unusual, and if the values below -1.0 the structure considered as highly unusual. The total average G-factor of the MurG model prior to optimization is -0.04 greater than the tolerable values of -0.5 so that even before minimization the modeled structure fulfill the criterion of the plot. On the other hand, the overall average G-factor of the minimized MurG as well as the template had 0.17 and 0.4 respectively which is far away the acceptable values of -0.5, which had a more reliable modeled structure (R. A. Laskowski et al., 1993; Roman A Laskowski et al., 1996) (Table 1). Before minimization, the MurG's structural quality factor value was 66.85% obtained from ERRAT, whereas, YASARA minimized MurG and template score value were 99.43% and 95.02% respectively. As a result, after minimization the ERRAT output indicated that the overall quality factor was increased significantly, which indicated that the predicted model was well optimized in terms of geometry and it is considered for further interaction analysis (Table 1). The outcome of the ERRAT stated as the proportion of the overall residues for which the calculated error score falls below the ninety percent rejection limit (Colovos & Yeates, 1993). The Verify_3D score of the MurG configuration before minimization revealed that 80.55% of the residues have averaged 3D-1D score >= 0.2, which is passed according to Verify_3D value analysis, on the other hand YASARA minimization MurG and template had 83.84% and 98.01% of the residues within the 3D-1D value >= 0.2, that was a sign of highly reliable modeled structure. Based on Verify_3D model evaluation analysis output greater than or equal eighty percent of the amino acid residues had scored more than 0.2 in the 3-D/1-D organization (Bowie et al., 1991; Luthy et al., 1992) (Table 1).

The YASARA energy minimization server works related to protein structure refinement problems through force-field in combination with the AMBER package (Cornell et al., 1995). This combination reached to all features of torsional potential and stereochemistry (Keedy et al., 2009) as well as diverse restraints to intensify the accuracy through refining physicochemical, and side-chain values and amending a certain non-bonded interactions, van der Waals repulsion energy and ions (Krieger et al., 2009). The total amount of energy and score values of the modeled structure was -34687.7 kJ/mol and -2.83 score values, respectively, whereas in the YASARA minimized modeled structure the energy decreased to -191344.9 kJ/mol and the stability increased to -0.54 score values. The increasing in the score values and the decreasing of the amount of energy in the minimized MurG was an indicator of structural stability.

3.3. Virtual high-throughput screening (vHTS)

During the discovery of potential drugs, one of the critical steps is high-throughput virtual screening, which is a computer-based skill, an initial and essential step for drug design and development processes. Structure based virtual screening is one of the most successful technique to identify the promising inhibitors against various disease causing proteins. One of the recent study reported, the structure based virtual screening approaches using ZINC database to identify the promising inhibitors against Microtubule affinity-regulating kinase 4 (MARK4). The over expression of this protein lead to several diseases particularly cancer, neurological disorders etc (Mohammad et al., 2019). A large dataset of compounds (virtual library) and the modeled structure were prepared to distinguish each ligand based on estimated binding affinity, score and pose to the target modeled structure catalytic site via virtual screening. On the basis of binding affinity value and docked conformation, these molecules were screened via PyRx 8.0 (Vina Wizard) tool. This means that the ligands were screened out according to binding free energy and binding. In the screening of ligands against MurG modeled structure, five thousand two hundred seventy nine (5279) compounds were obtained with a good binding affinity score towards the binding site of the modeled structure from thirty thousand seven handed ninety two (30,792) ligand library which could be additionally screened for searching of a possible inhibitors of MurG. Interestingly, recent study reported that virtual screening of 104 herbal molecules was performed against one of the Mur protein involved in peptidoglycan pathway namely MurA from MDRAB. They utilized standard antibiotics as a control and compared with lead molecules and they identified promising inhibitors via combined *in silico* and *in vitro* studies (Skariyachan et al., 2019). In our present study we also used the two antibiotic molecules (Clinafloxacin and Imipenem) which are inhibiting the growth of the MDRAB. These can be used as a control molecules for comparing the results obtained from virtual screening in terms of estimated free energy of binding, estimated inhibition constant, intermolecular interactions, etc.

3.4. Hits selection and pharmacokinetics analysis

The controls used in this study showing the estimated free energy of binding of -6.9 in case of Clinafloxacin and -5.6 in case of Imipenem. This indicate that in comparison with control molecules, the results obtained from virtual screening showing better binding affinity towards MurG protein. In the study, ligand shows the binding affinity cut-off < -8.7 kcal/mol towards MurG considered for further screening whereas remaining molecules which does not satisfy the cut-off excluded for further studies. One of the recent study also utilized binding affinity cut-off as one of the important filtering parameter for identifying the best hits after high throughput virtual screening and docking studies (Mohammad et al., 2019). As a result, eighty six (86) hits were obtained out of five thousand two hundred seventy nine (5279) natural compounds. From the beginning, the screening techniques as well as ligand selection were done based on structure based drug design method. The natural compounds which are satisfying our cut-off value of binding affinities less than -8.7 kcal/mol were filtered based on their free energy of binding, and other parameters. Twelve compounds from eighty six (86) hits were passed all the filters and presented in tabulated form to show the binding affinities with the inhibition constant (Table 2), the physicochemical properties (Table 3), and pharmacokinetics analysis results (Table 4 and 5). The selected compounds were further screened to remove undesirable ligand features to drug-likeness according to the physicochemical properties apart from the binding affinity score, which indicate that low permeability and absorption are taking place when a molecule/compound violates a Lipinski rule. Consequently, 40 ligands from 86 hits were fitted for physicochemical properties filters using DataWarrior (Veber et al., 2002) and PreADMET (Lee et al., 2004) Tools. The hydrogen bond donors below 5, hydrogen bond acceptors below 10, molecular weight below 500 Da, an average of clogP values below 5, rotatable bonds score below 10 and satisfied drug-likeness were the physicochemical properties which were used to test each natural ligand for the fulfillment of drug-likeness features. Physicochemical properties of finally selected 12 ligands are shown in Table 3.

As indicated in the above, forty ligands were selected after screening using physicochemical properties. The result indicated that, these compounds are the possible lead molecules which may cure infections caused by *A. baumannii* after checking pharmacokinetics and toxicity properties. On the other hand, as a result of poor pharmacokinetics and toxicity difficulties, mostly drug-able molecules fail to pass quality standards. Therefore, ligands that fitted with physicochemical properties were tested for ADMET parameters. ADMET stands for absorption, distribution, metabolism, excretion, and toxicity. These properties are easily identified ligands which are pharmaceutically appropriate or acceptable compounds for the treatment of *A. baumannii* infections. The ADME property consists of Human Intestinal Absorption (HIA), Blood-Brain Barrier penetration (BBBP), Renal Organic Cation Transporter (ROCT), Aqueous Solubility, Cytochrome P450 (CYP450 2D6), Polar surface area (PSA), Water solubility (cLogS) and Relative Polar surface area (RPSA) (Table 4). Similarly, the Toxicity properties include AMES (Test) Toxicity, Carcinogenicity, Mutagenicity, Tumorigenicity, Reproducibility and Irritant (Table 5). The ADME parameters revealed that, all the natural compounds had the power to penetrate blood-brain barrier and absorb in the human intestine as well as near to the optimum polar surface area and water solubility values (Shen, Cheng, Xu, Li, & Tang, 2010), on the other hand, all the designated compounds does not had the ability to impede Cytochrome P450 (CYP450 2D6) and Renal Organic Cation Transporter. This results indicated that, according to the model calculations, all the selected compounds were in agreement with the standards (Cheng et al., 2011) except ZINC12602407 and ZINC08925939 respectively. On the other hand, the toxicity results explained that, of all the molecules, these two compounds, namely ZINC15676070 and ZINC15675820 showing mutagenicity as well as tumorigenicity. Similarly, ZINC08779971 and ZINC12603201 were responsible for irritation. In addition to this, of all the ligands (forty) twenty five showed AMES Toxic but positive results for Carcinogens. Overall, according to the ADMET results, twelve (12) compounds, including ZINC08857096, ZINC15707240, ZINC09186673, ZINC08856120, ZINC08439415, ZINC22920899, ZINC12530134, ZINC39960784, ZINC15675806, ZINC15675922, ZINC22936720 and ZINC15675812 had fulfilled the ADMET properties. These ligands were suggested as a putative inhibitors, which may inhibit the function of MurG in *A. baumannii* (Tables 4 and 5). HTVS was already performed against MurG of *Mycobacterium tuberculosis* and MurA of MDRAB to identify the best molecules on the basis of binding affinity, estimated inhibition constant, binding mode, docking orientation, interacting residues and intermolecular interactions as well as drug-likeness properties and ADMET parameters. As previous studies (Saxena et al., 2018; Skariyachan et al., 2019), we observed some of the promising molecules which are also showing estimated inhibition constant at nM level against MurG and satisfying all the parameters.

3.5. Structure analysis of MurG-ligand complexes

The structure of MurG has two domains separated by a deep cleft, namely *N* (domain I) and *C*-(domain II) terminal domains containing a long chain of three hundred sixty five (365) amino acid residues. In addition to this, it has been critical residues which are very important for the substrate binding site. Both control as well as molecules obtained from HTVS showing the interactions with several common residues towards active site of MurG. The crucial amino acid residues used for ligand interaction include Gly20, Arg170, Gly200, Ser201, Gln227, Phe254, Leu275, Thr276, and Glu279. Accordingly, knowing the above stated residues are used to target for drug to hinder the normal role of MurG. The outcomes of interaction analysis indicated that MurG and the ligands undertaken numerous non-covalent interactions. The foremost interactions in between the selected ligands and the MurG structure include, hydrogen bonding and pi-effects (pi-pi, pi-alkyl, alkyl, pi-sulphur etc.) as well as several hydrophobic and van der Waals interactions. This indicated that each natural ligands interact with the MurG around the catalytic site region with substantial binding affinity. Of all the twelve designated ligands, ZINC08857096 and ZINC15707240 had the lowest binding energy (good affinity towards MurG) (─9.6 kcal/mol) whereas ZINC15675812 (─8.7 kcal/mol each) has been comparatively a lowest affinity towards the MurG in terms of number as well type of intermolecular interactions. These compounds were bounded between the *N*-terminal and the *C*-terminal domains of the MurG model. The following ligands namely ZINC08857096, ZINC15707240 and ZINC15675812 were stabilized by six, two and two hydrogen bonds, pi-interactions with five residues of MurG for each ligand (Table 6) as well as numerous hydrophobic and van der Waals interactions respectively (Supplementary information-Figure 6). The amino acid residues of MurG contributed in hydrogen bond formation with the different natural ligands were Leu275, Thr276, Gly273, Gly20, Gly21, His299, Ala274, Gln300, Thr19, Arg271 and Asn131 (Table 6). Similarly, Val198, Arg170, Phe24, Leu275, Arg271, Ala274, Cys270, Met258, His299, and His22 were amino acid residues underwent in pi-interactions with the selected ligands (Table 6). On the other hand numerous residues were also contributed for the formation of hydrophobic and van der Waals interactions with each ligand (Supplementary information-Figure 6 and 8). The unique stability of the selected ligands towards the binding site of MurG modeled structure is enhanced by high number of hydrogen bonds, pi effects, hydrophobic and van der Waals interactions. Most of the studies reported the best lead molecules on the basis of presence of aforementioned intermolecular interactions between active site of protein and ligand molecules (Mohammad et al., 2019; Saxena et al., 2018; Skariyachan et al., 2019). Out of twelve compounds, four were selected (Figures 3-6) for further MD simulations to understand the structural stability of these four molecules towards MurG structure.

The ligand ZINC08857096 is forming hydrogen bonds with Leu275 (Distance=2.37 Å), Thr276 (two times) (Distance=2.47 Å and 2.99 Å), Gly273 (Distance=2.85 Å), Gly20 (Distance=2.80 Å) and Gly21 (Distance=2.92 Å) of MurG (Figure 3a), and pi-interactions with Val198 (Distance=5.12 Å), Arg170 (Distance= 5.10 Å), Phe24 (Distance= 3.76 Å), Leu275 (Distance= 4.43 Å) and Leu275 (Distance= 2.62 Å) (Figure. 3b). In addition to this, the ligand also show van der Waals interactions with numerous residues of MurG such as Gly200, Arg271, Leu292, Gln300, Thr19, His22, Asn131, His299, Ala274, Gly21 Gly20 and Glu279 (Figure 3c) and hydrophobic interactions with a number of residues of MurG including Asn131, Gly21, Gly20, Phe24, Glu279, Arg271, Gly273, Ala274 and His299 (Table 6). These interaction and bond formation indicated that there is a substantial binding affinity of ligand ZINC08857096 with the MurG as a result of a significant amount of hydrogen bonds as well as pi, hydrophobic and van der Waals interactions which is supported by recent study reported on MARK4 protein-ligand (ligands identified through HTVS against ZINC database) interactions (Mohammad et al., 2019).

In the same manner the ZINC09186673 form hydrogen bond with Thr276 (Distance=2.53 Å), Thr276 (Distance=2.44 Å), Ala274 (Distance=2.78 Å), Leu275 (Distance=2.68 Å), Gly20 (Distance=3.48 Å) (Figure 4a) and pi-interactions with Leu275 (Distance=5.11 Å), Leu275 (Distance=3.96 Å), Leu275 (Distance=3.82 Å), Arg271 (Distance=5.35 Å), Arg271 (Distance=3.61 Å), and Cys270 (Distance=4.73 Å) (Figure 4b). On the other hand the compound interacting with a number of MurG residues in the form of van der Waals (Phe24, Thr19, Gly21 Asn131, Gly273, Gln300, and Glu279) (Figure 4c) and hydrophobic (Glu279, Phe24, Leu275, Gly20, Asn131, Ala274, Gly21, Gly273, and Arg271) interactions (Table 6). In these association the result output indicated that there is a strong interaction between MurG and the ZINC09186673 but it is a slightly different in terms of intermolecular interactions and interacting residues in comparison with ZINC08857096. The number and type of interactions are in line with recent study reported about natural ligands which showed promising interaction towards modeled structure of MurA from MDRAB (Skariyachan et al., 2019).

In the same way the ligand ZINC08856120 making hydrogen bonds with Thr276 (Distance=2.48 Å), Gln300 (Distance=3.03 Å), Leu275 (Distance=2.32 Å), Thr19 (Distance=2.59 Å) and Gly20 (Distance=2.70 Å) (Figure 5a) and pi-interactions with Val198 (Distance=5.05 Å), Leu275 (Distance=2.64 Å), Leu275 (Distance=4.51 Å) and Phe24 (Distance=3.64 Å) (Figure 5b). In addition to these, the ligand form van der Waals interactions with Arg170, Gly200, Arg271, Ala274, Gly273, His299, Asn131, and Gly21 residues of MurG (Figure 5c) and hydrophobic interaction with Gly200, Gly21, Arg271, Gln300, Gly273, His299, Asn131, Ala274, His22, Thr19, Gly20, and Phe24 residues of the modeled structure (Table 6). These interactions showed that the affinity of the ligands towards the target structure is a cumulative effect of the type and number of interactions as well as type and number of residues which were participated throughout the interaction. A higher number of intermolecular interactions formed between MurG and the ligands indicated that ligands showing strong and significant binding affinity towards active site of MurG. Recent report related to MurG from *Mycobacterium tuberculosis* showed that higher number of hydrogen bonds towards ligand molecules are an indication of significant interactions (Saxena et al., 2018).

Finally, the ligand ZINC08439415 also making hydrogen bonds with Thr276 (Distance=2.45 Å), Thr276 (Distance=3.06 Å), Leu275 (Distance=2.43 Å), Gly21 (Distance=2.96 Å) and Gly20 (Distance=2.79 Å) (Figure 6a) and pi-interactions with Leu275 (Distance=2.64 Å), Leu275 (Distance=4.41 Å), Phe24 (Distance=3.76 Å), Arg170 (Distance=5.03 Å), Val198 (Distance=4.84 Å), Cys270 (Distance=4.16 Å), Arg271 (Distance=4.58 Å) and Arg271 (Distance=3.88 Å) (Figure 6b). Here also numerous MurG residues have been formed van der Waals (Gly273, Gln300, Asn131, Thr19, Ala274, Glu279, Gly199, and Gly200) (Figure 6c) and hydrophobic (Leu275, Glu279, Phe24, Gly20, Val198, Gly273, Gly200, Ala274, Cys270, Gln300, Gly199, Arg271, and Gly21) interactions with the indicated ligand (Table 6). Hence, a number of the hydrogen bond, pi, hydrophobic and van der Waals interactions are a sign of significant binding affinity among MurG and the ligand. Generally, this results indicated that the compound that had the highest binding affinity which made hydrogen bond as well as pi, hydrophobic and van der Waals interaction with MurG catalytic site ceased the process of cell wall synthesis, hence, hinder the *A. baumannii* growth. Overall various *in silico* studies of drug target proteins indicated that higher number hydrogen bonds, pi-bonds, hydrophobic and van der Waals interaction are pivotal for determining the binding of the ligands towards protein (Mohammad et al., 2019; Saxena et al., 2018; Skariyachan et al., 2019).

Ligands are mostly interacting with positive charged amino acid residues of MurG model with the results obtained from analysis of electrostatic surface potential (Figure 7). From the results of virtual screening and docking, we identified four promising molecules which further subjected into MD simulation to identify the better anti-bacterial candidate among four for future experimental investigation.

3.6. Molecular dynamic simulation

Major limitation of docking calculation is it considered protein as rigid (some residues can be treated as flexible not entire protein) and ligand as flexible entities because of the computational complexity. In order to mimic the physiological function of protein and protein-ligand complex in living system, both should be considered as flexible and they are interacting in aqueous environment (Jameel et al., 2017; Mohammad et al., 2019). Now a days MD simulation is very effective computational tool to understand the structural stability and flexibility of bio-molecular complexes (Protein-Protein, Protein-Ligand, Protein-Carbohydrate, Protein-DNA and Protein-Lipid) in a regular time-scale in aqueous environment by analyzing various structural parameters such as Root Mean Square Deviation (RMSD), Root Mean Square Fluctuation (RMSF), Radius of Gyration (RoG), Inter-molecular hydrogen bonds, Solvent Accessible Surface Area (SASA) and Essential Dynamics (ED). From results of docking and virtual screening analysis, we identified four promising molecules (ZINC08857096, ZINC09186673, ZINC08856120 and ZINC08439415). Here, we used free form of MurG and four ligand bounded forms of MurG for MD simulation studies. Furthermore, in order to understand the conformational stability, dynamics and structural stability of protein–ligand complex, a long range MD simulation of 100 ns was performed on best four MurG-Ligand (MurG-ZINC08857096, MurG-ZINC09186673, MurG-ZINC08856120 and MurG-ZINC08439415) complexes and free form of MurG.

3.6.1. Root Mean Square Deviation (RMSD)

RMSD is used to measure the structure stability of protein or protein-ligand complexes in regular time scale (Kuzmanic & Zagrovic, 2010). Here, we observed minor structure deviations in both protein and in all protein-ligand complexes until 30ns. After that, both protein and protein-ligand complexes started to attain the stabilization state and it maintains the stability until end of 100 ns simulation period (Figure 8a). The average RMSD values of MurG, MurG-ZINC08857096, MurG-ZINC09186673, MurG-ZINC08856120 and MurG-ZINC08439415 are 2.60 Ǻ, 3.18 Ǻ, 2.97 Ǻ, 3.48 Ǻ and 3.45 Ǻ respectively. Overall results explained that all four ligand molecules does not significantly influences the structural stability of MurG. However they are showing nearly similar kind of RMSD pattern particularly ZINC09186673, and we observed slight variation in comparison with ligand free form of MurG. The similar kind of pattern explained that they might share common interacting site of MurG and they could be starting model for synthesis and identification of more promising molecules against MDRAB. Together all four trajectory complexes at 100ns time period, the molecules ZINC09186673 and ZINC08857096 interact strongly than other two inhibitors. These results were corroborate with previous docking calculation. Based on the average RMSD values obtained from this analysis, the structural stability for MurG-ZINC09186673 complex is closer to ligand free form of MurG. This hypothesis will be further validated by RMSF analysis.

3.6.2. Root Mean Square Fluctuation (RMSF)

RMSF is another essential structural parameter to identify the flexible and non-flexible (rigid) region of protein. Moreover, it also identified the flexible residues in the protein and thus allow us to explore the conformational flexibility of the protein structure (N. Kumar, Srivastava, Prakash, & Lynn, 2019; Mohammad et al., 2019). In this study, we observed higher fluctuations in most of the places of protein structure particularly the region in between *C* and *N* terminal domains because of the presence of several loops and coil like secondary structural elements (Figure 8b). One of the complex, MurG-ZINC08439415 is showing larger fluctuations (average RMSF 17.55 Ǻ) in comparison with other protein-inhibitor complexes. Except this, the remaining are showing nearly similar RMSF pattern and they matched with previous RMSD analysis. The average RMSF values of MurG, MurG-ZINC08857096, MurG-ZINC09186673, MurG-ZINC08856120 and MurG-ZINC08439415 are 9.55 Ǻ, 13.11 Ǻ, 8.99 Ǻ, 11.77 Ǻ and 17.55 Ǻ, respectively. Additionally, the fluctuations were found to be minimized upon binding to some of compounds, particularly ZINC09186673.This will additionally supported that this molecule bind strongly towards binding site of MurG model.

3.6.3. Radius of Gyration (RoG)

Radius of Gyration is an useful tool to understand the compactness and folding properties of proteins and also explore the influence ligand towards conformational changes of protein structure (Prakash, Kumar, Lynn, & Haque, 2019). In this study, in comparison with ligand free MurG form, the complex structure showing higher RoG values throughout 100 ns time period which revealed that the binding of these molecules with protein might have caused an alteration, thus lead to conformational changes of the protein structure (Figure 8c). The average RoG values for free MurG, MurG-ZINC08857096, MurG-ZINC09186673, MurG-ZINC08856120 and MurG-ZINC08439415 were found to be 22.23 Ǻ, 23.32 Ǻ,23.28 Ǻ, 22.93 Ǻ and 23.58 Ǻ, respectively. Among four molecules, the RoG values of MurG-ZINC08856120 and MurG-ZINC09186673 complexes are closer to ligand free form of MurG which conclude that these are significantly binding to the protein and they maintain the compactness which do not affect the folding properties of MurG. In contrast, another molecule ZINC08439415 showing higher RoG values which may leads to high structural fluctuations as RMSD and RMSF results.

3.6.4. Hydrogen bond and Solvent Accessible Surface Area (SASA) analysis

Understanding the formation and deformation of hydrogen bonds between protein and ligand molecules is an important in MD simulation to understand the binding efficacy of putative inhibitors obtained from HTVS and docking analysis. Gain and lose of hydrogen bonds between protein and ligand helped us to identify the binding potential and interacting nature of the ligand molecules towards drug target protein (N. Kumar et al., 2019). During this analysis, we found that 6-7 average hydrogen bonds between active site of MurG and four ligand molecules. The number intermolecular hydrogen bonds for MurG-ZINC08857096, MurG-ZINC09186673, MurG-ZINC08856120 and MurG-ZINC08439415 are 6 (Asn131, Ser201, Arg271, Leu275, Thr276 and Gln300), 7 (Thr19, Gly20, Gly21, Arg271, Thr276, H299 and Gln300), 7 (Arg271, Ala274, Thr276 (forms 2 hydrogen bonds), Leu275 and Gln300), and 7 (Asn131, Ser201, Leu201, Gly203, Arg271 and Gln300) respectively. During these simulations, four ligand molecules (Figure 9a-d) are frequently interacting with active site residues of MurG through hydrogen bonds particularly Arg271 and Gln300 are mostly interacting residues in all the molecules which are corroborating with molecular docking results.

Moreover, we performed SASA analysis of ligand free MurG and four ligand bounded MurG complexes (MurG-ZINC08857096, MurG-ZINC09186673, MurG-ZINC08856120 and MurG-ZINC08439415) (Figure 8d). This analysis is mainly used to understand the solvent behavior (Hydrophilic or Hydrophobic) of the protein and also study the influence of ligand binding towards solvent effects of protein (Mazola et al., 2015). There is no major changes observed in SASA profile of MurG and two MurG-ligand complexes (ZINC09186673 and ZINC08856120) whereas slight changes were observed in other two ligand bounded complexes namely MurG-ZINC08439415 and MurG-ZINC08857096. The average SASA values for free MurG, MurG-ZINC08857096, MurG-ZINC09186673, MurG-ZINC08856120 and MurG-ZINC08439415 were found to be 17664.79 Ǻ, 18801.37 Ǻ, 18414.32 Ǻ, 18385.26 Ǻ and 18774.25Ǻ, respectively. Overall results explained that the MurG is stable after the binding of ZINC09186673 and ZINC08856120 molecules to the active site.

3.6.5. Principal component (Essential dynamics) analysis

The principal component analysis was utilized to capture the essential and biological significant and relevant motions from the global trajectories of unbound MurG and MurG-ZINC08857096, MurG-ZINC09186673, MurG-ZINC08856120 and MurG-ZINC08439415 complexes using Essential Dynamics (ED) approach (Figure 10a-d). ED analyzes the collective fluctuations of the most variable regions of the MurG into two variables namely PC1 (Principal Component 1) and PC2 (Principal Component 2) which describes the majority of the fluctuations or movements observed in MD simulation (N. Kumar et al., 2019). PCA graph of ligand free MurG with ZINC08439415 bounded MurG explained that the protein is highly flexible in presence of this particular compound. In contrast other molecule, ZINC09186673 is significantly enhances the stability of MurG because it covers the lesser region in conformational space of two principal components. This results along with previous analysis such as RMSD, RMSF, and RoG additionally supported that this molecule could be most promising one which inhibit the function of MurG. As MurG-ZINC09186673 complex, ZINC08856120 is rated as second best which also maintain the protein compactness. ZINC08439415 molecule is alter the protein structure into expanded conformation which may be reason for covering huge area in conformational space which are corroborate with all our analysis. After observing RMSD, RMSF, RoG and PCA results, we proposed that ZINC09186673 and ZINC08856120 molecules could be the most promising molecules for inhibiting the function of MurG in *A. baumannii*.

3.6.6. Comparative analysis of Pre and Post MD Simulation

Structural superposition before and after MD simulation of ligand free MurG structures showed RMSD of 2.1 Ǻ (Figure 11a). This indicate that the MD simulated MurG structure is reasonable and well geometrically optimized which does not deviate in comparison with initial structure. Moreover we also performed same pattern of structural superposition analysis for four ligand bounded MurG structures (Figure 11b-e) which explain that the RMSD between before and after MD simulation of following two complexes (MurG-ZINC09186673 and MurG-ZINC08856120) are significant with the values of 2.31 Ǻ and 2.30 Ǻ, respectively. This is additionally supported that these two ligand molecules are oriented in the active site of MurG throughout the MD simulations which corroborate with previous MD analysis. However the orientation of other two ligand molecules in the active site of MurG are slightly changed during MD simulation especially ZINC08439415 (RMSD 3.01 Ǻ). Similar kind of observations are noticed in all the MD analysis of MurG with ZINC08439415 and ZINC08857096 complex which revealed that these two molecule are not considered for further studies.

4. Conclusion

In-silico based exploring of potential lead candidates from the library of natural products for the treatment of *A. baumannii* infections is accomplished in the present study with the aid of systematic computational analysis. Initially we predicted 3D model of MurG using comparative protein modeling approach based on homologous MurG structure from *Escherichia coli* (Strain K12) (PDB ID: 1F0K). After this, we performed concordance binding site analysis to identify the putative ligand binding sites of optimized MurG model for virtual screening and docking studies against natural compounds subset of Zinc database. After the thorough examination of drug-like, pharmacokinetic properties and interaction mechanism, we shortlisted four promising molecules for MD simulation. With the obtained results of RMSD, RMSF, RoG, Hydrogen Bonds and Essential dynamics analysis, these following two molecules (ZINC09186673 and

ZINC08856120) are worth for further investigation. An identification of potential anti-bacterial lead candidates is really challenging and cumbersome process in terms of time, experimental cost and manpower. Although the presented computational analysis will be useful to identify the best lead molecules, further experimental studies is always needed for validating the results obtained from computational screening which is one of limitation of this kind of study. However this *in silico* studies saves time and experimental duration. The proposed molecules obtained from this analysis is beneficial to the multi-disciplinary scientific groups for further experimental investigation.

Conflict of interest

None

Ethical standards

Ethical standards is compulsory for studies relating to human and animal subjects

Acknowledgments

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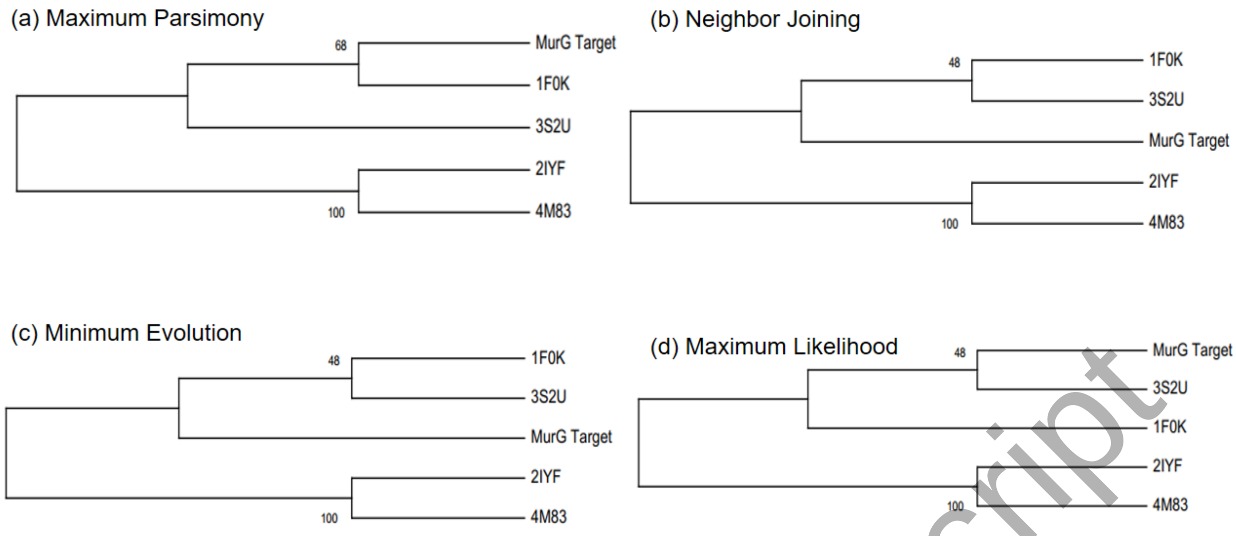


Figure 1. The phylogenetic trees of MurG sequence from *A. baumannii* with the UDP-N-acetylglucosamine-N-acetylmuramyl (pentapeptide) pyrophosphoryl-undecaprenol N-acetylglucosamine transferase from *Escherichia coli* Strain K12 (PDB accession No: 1F0K), *Pseudomonas aeruginosa* (strain ATCC 15692) (PDB accession No: 3S2U), as well as Oleandomycin glycosyltransferase from *Streptomyces antibioticus* (PDB accession Nos: 2IYF, 4M83*)*.

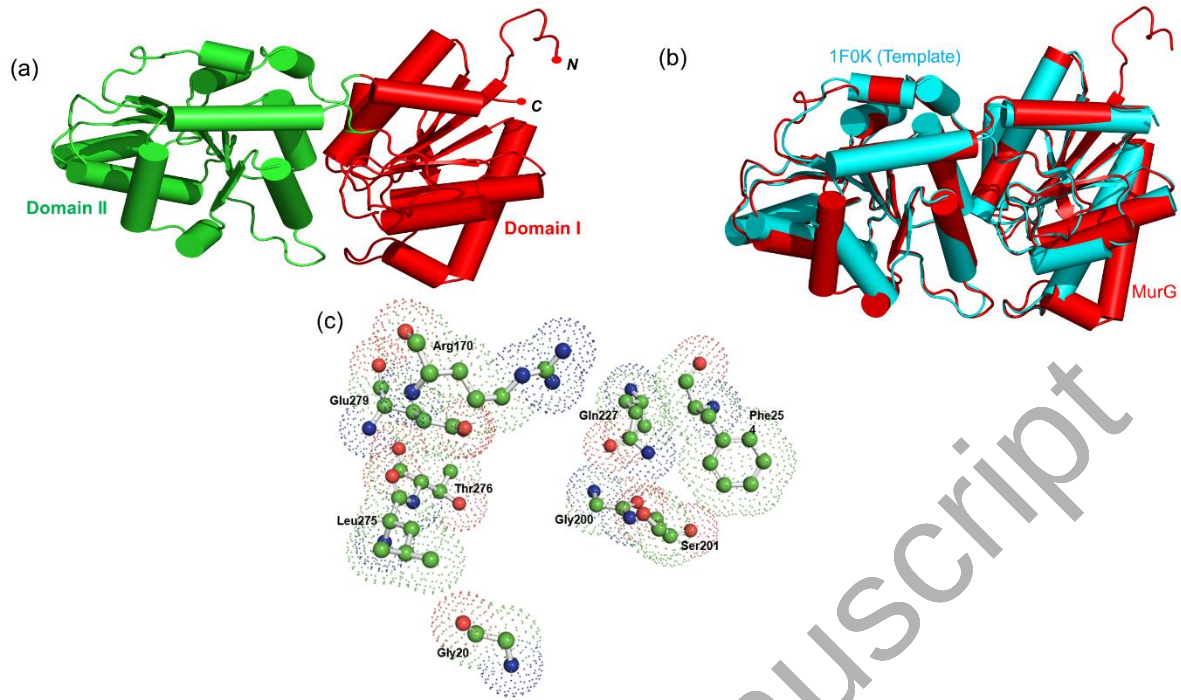


Figure 2. The 3D predicted model of MurG (red) with template 1F0K (cyan). (a) Predicted model of MurG with domains (domain I and domain II), (b) Superposition of 1F0K (cyan) with MurG (red) (RMSD = 0.475 Å), (c) Amino acid residues involved in UDP-GlcNAc binding pocket (Gly20, Arg170, Gly200, Ser201, Gln227, Phe254, Leu275, Thr276, and Glu279).

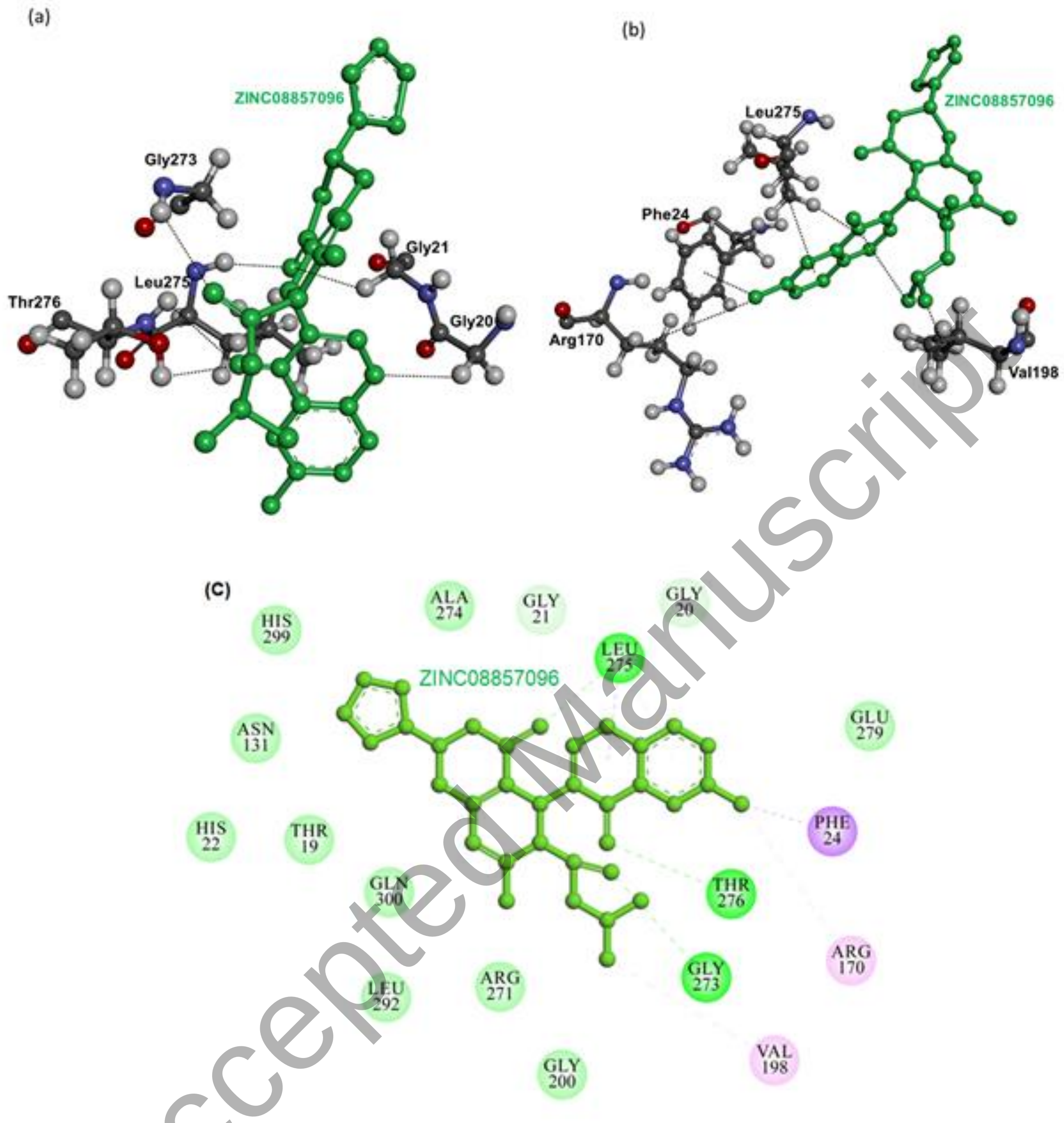


Figure 3. Interactions of ligand ZINC08857096 to the MurG (a) three-dimensional orientation of catalytic site residues of MurG interacting with compound ZINC08857096 (green) through hydrogen bond (b) three-dimensional orientation of catalytic site residues of MurG interacting with ligand ZINC08857096 (green) via pi-interaction (c) two-dimensional representation of MurG residues interacting with ligand ZINC08857096 through van der Waals interactions (slightly green color residues), hydrogen bond (deep green color residues) and pi-interaction (pink color residues).

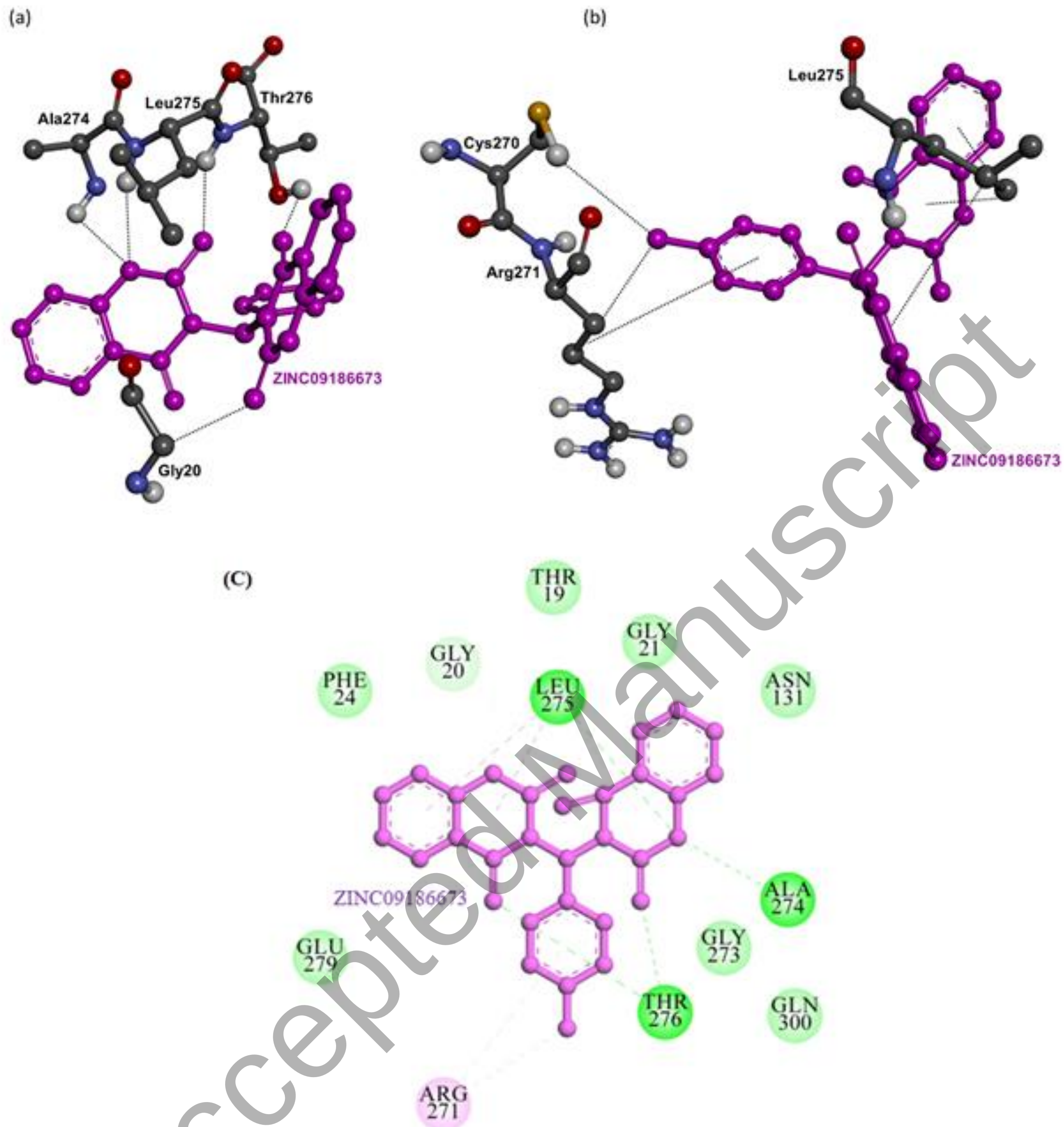


Figure 4. Interactions of ligand ZINC09186673 to the MurG (a) three-dimensional orientation of catalytic site residues of MurG interacting with compound ZINC09186673 (magenta) through hydrogen bond (b) three-dimensional orientation of catalytic site residues of MurG interacting with ligand ZINC09186673 (magenta) via pi-interaction (c) two-dimensional representation of MurG residues interacting with ligand ZINC09186673 by van der Waals interactions (slightly green color residues), hydrogen bond (deep green color residues) and pi-interaction (pink color residues).

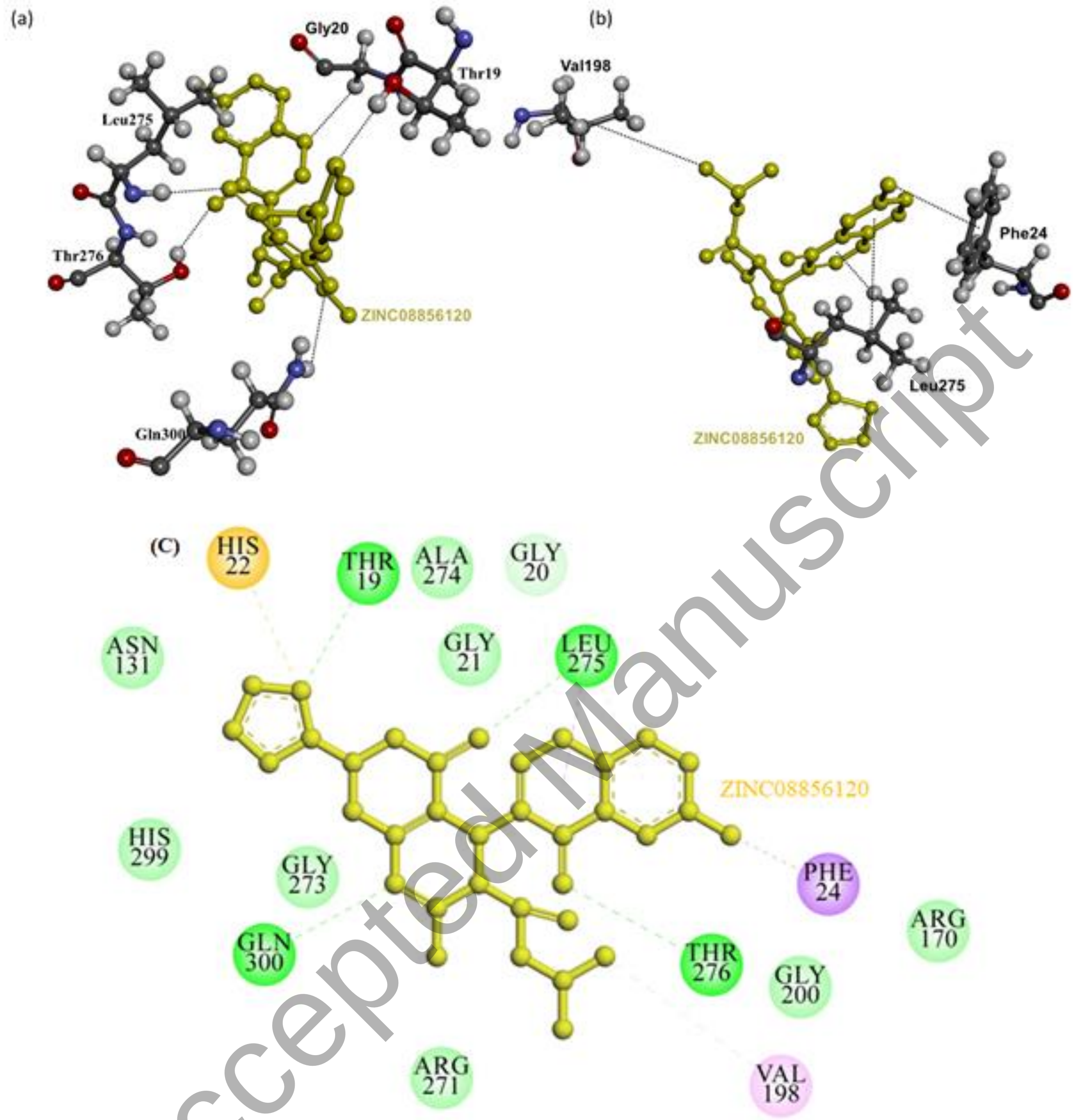


Figure 5. Interactions of ligand ZINC08856120 to the MurG model (a) three-dimensional orientation of catalytic site residues of MurG interacting with compound ZINC08856120 (yellow) through hydrogen bond (b) three-dimensional orientation of catalytic site residues of MurG interacting with ligand ZINC08856120 (yellow) via pi-interaction (c) two-dimensional representation of MurG residues interacting with ligand ZINC08856120 by van der Waals interactions (slightly green color residues), hydrogen bond (deep green color residues) and pi-interaction (pink and yellow color residues).

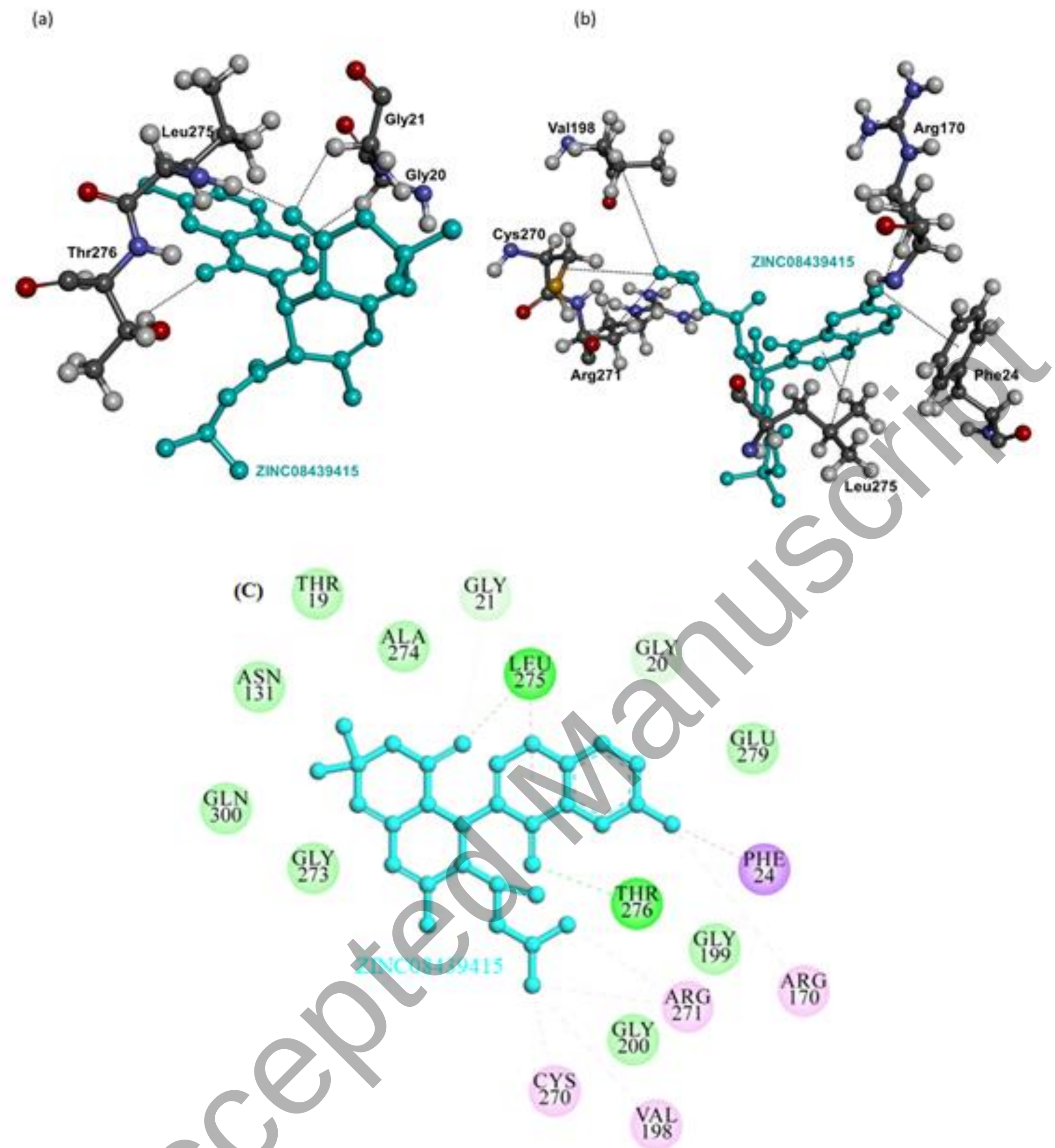


Figure 6. Interactions of ligand ZINC08439415 to the MurG (a) three-dimensional orientation of catalytic site residues of MurG interacting with compound ZINC08439415 (cyan) through hydrogen bond (b) three-dimensional orientation of catalytic site residues of MurG interacting with ligand ZINC08439415 (cyan) via pi-interaction (c) two-dimensional representation of MurG residues interacting with ligand ZINC08439415 via van der Waals interactions (slightly green color residues), hydrogen bond (deep green color residues) and pi-interaction (pink color residues).

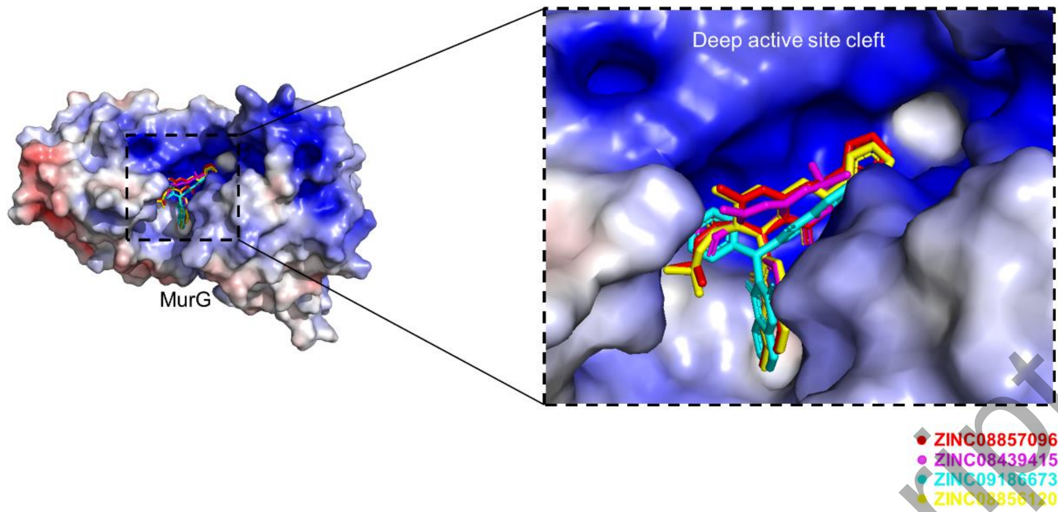


Figure 7. The electrostatic surface potential interaction of four selected ligands bounded with MurG model. Active site cleft are illustrated as zoom view.

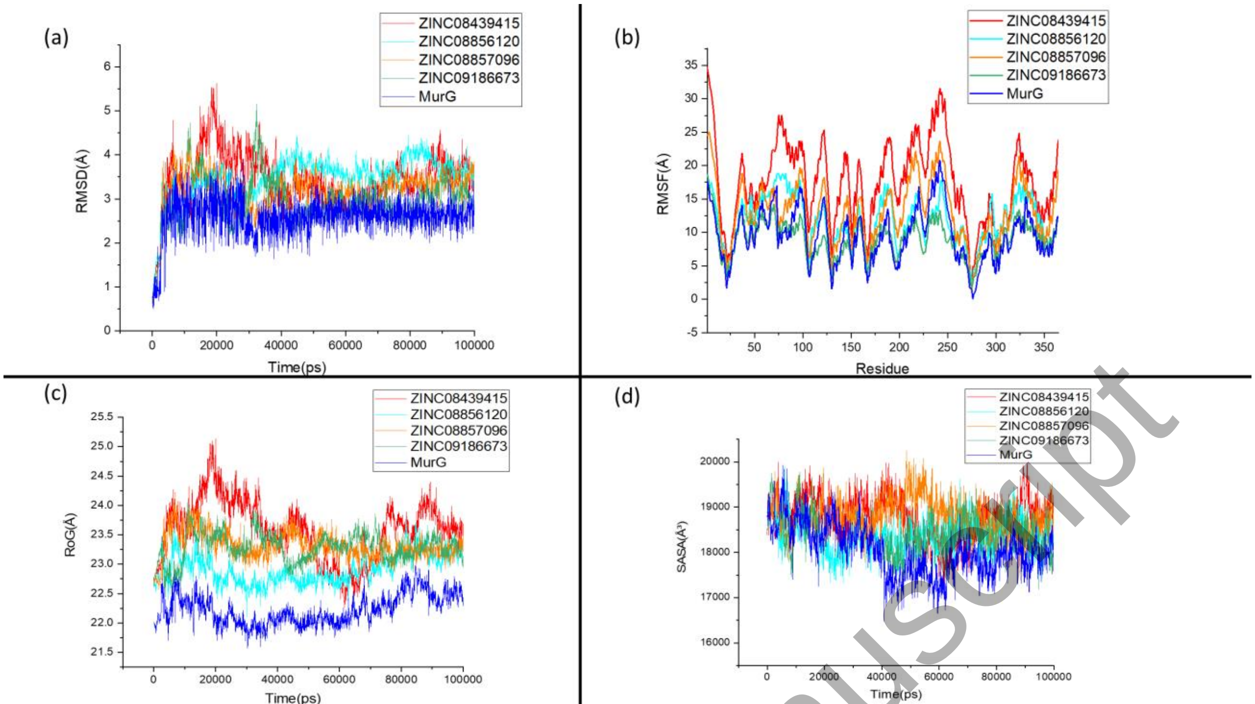


Figure 8: Molecular Dynamics Simulation results of Ligand free MurG and four ligand bounded MurG complexes. (a) Root Mean Square Deviation, (b) Root Mean Square Fluctuation, (c) Radius of Gyration and (d) Solvent Accessible Surface Area.

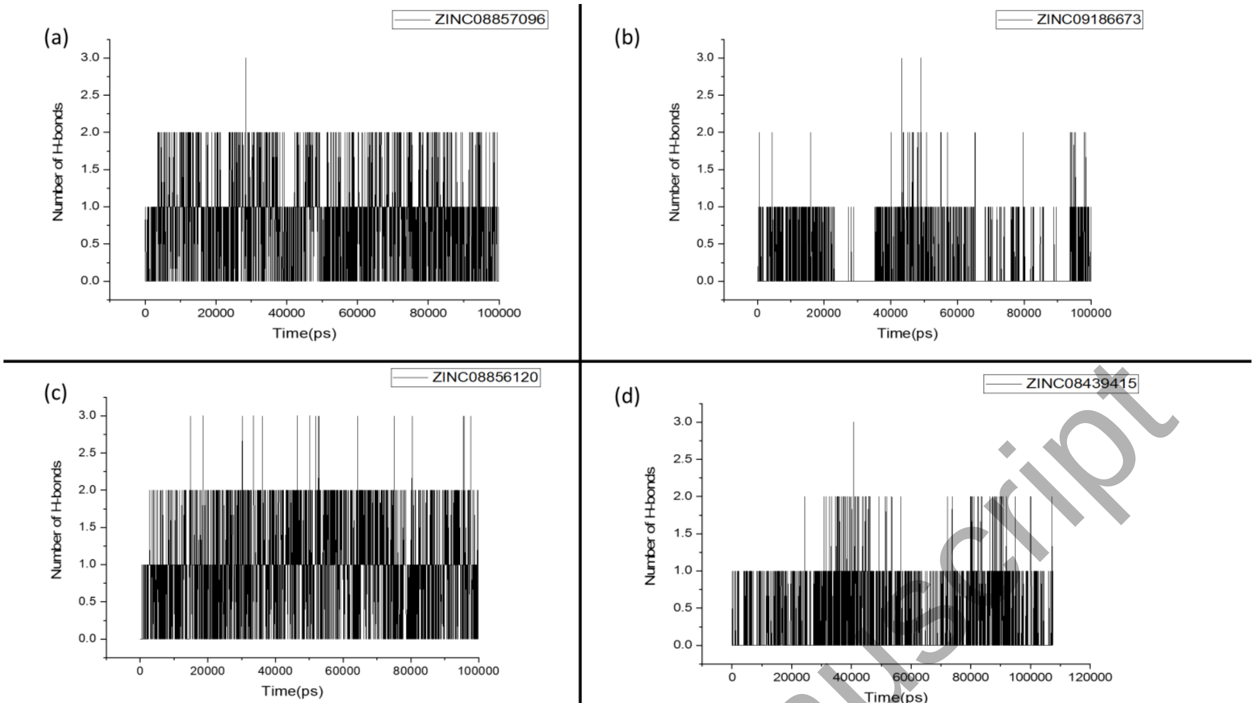


Figure 9: The number of hydrogen bonds are obtained from MD simulation for (a) MurG-ZINC08857096, (b) MurG-ZINC09186673, (c) MurG-ZINC08856120 and (d) MurG-ZINC08439415

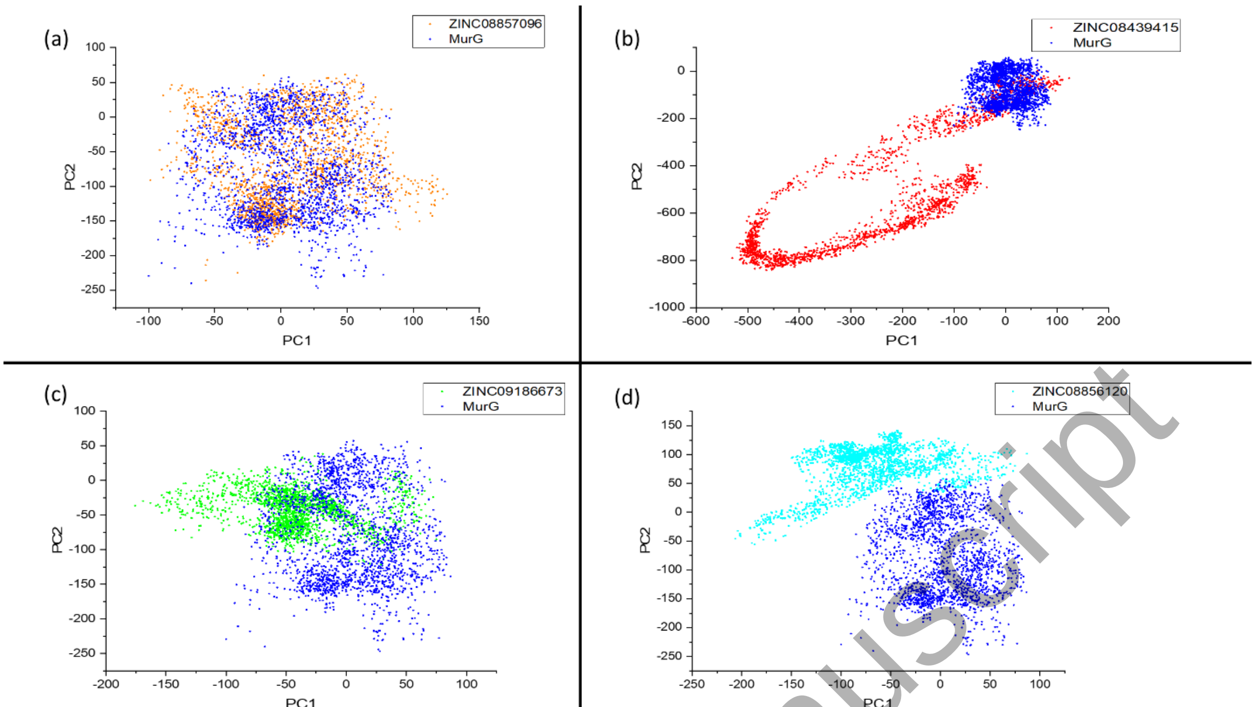


Figure 10: Essential Dynamics results (a) Ligand Free MurG with MurG-ZINC08857096 complex, (b) Ligand Free MurG with MurG-ZINC08439415 complex, (c) Ligand Free MurG with MurG-ZINC09186673 complex and (d) Ligand Free MurG with MurG-ZINC08856120 complex

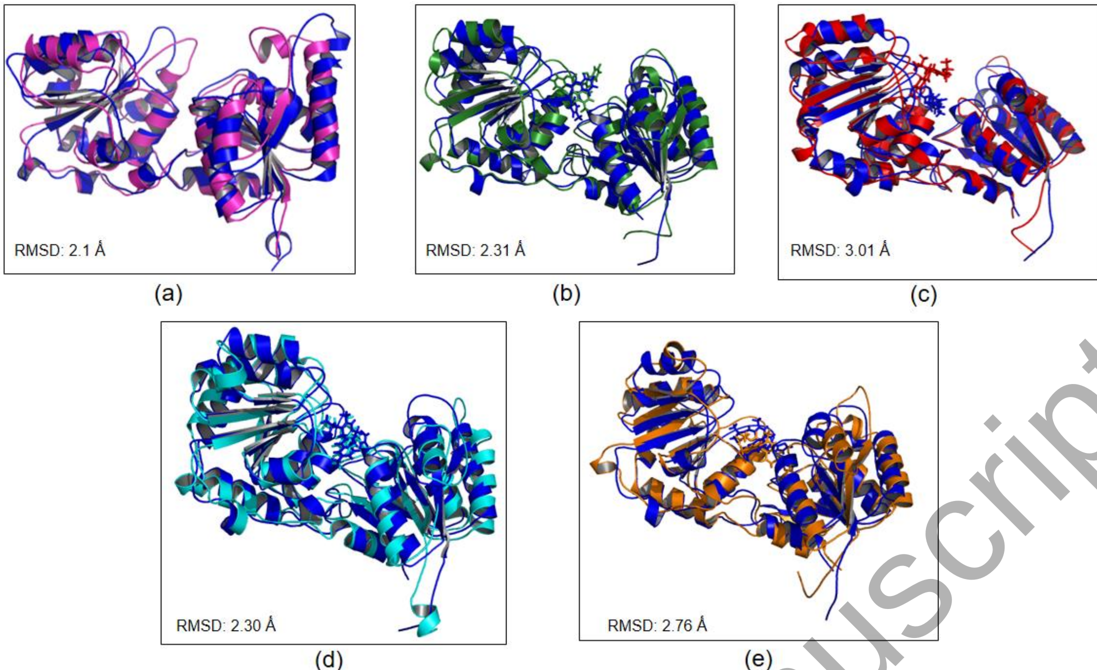
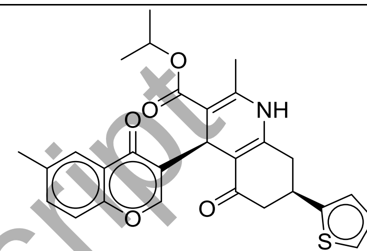
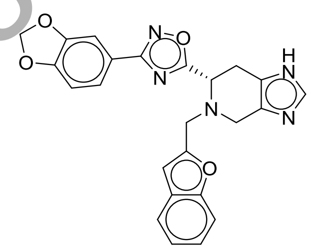
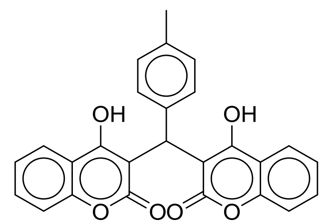
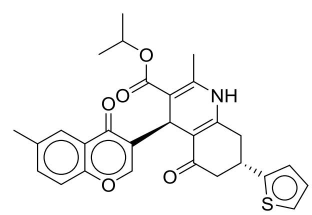
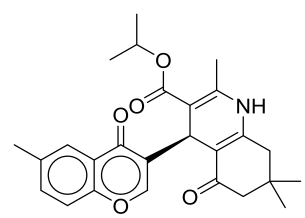


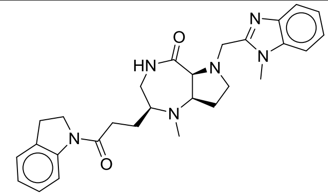
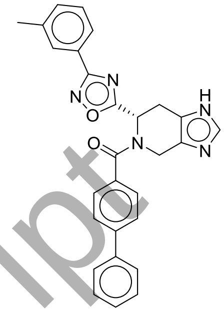
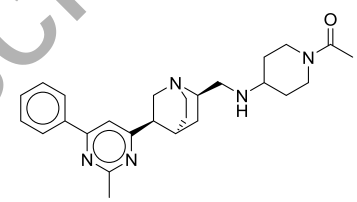
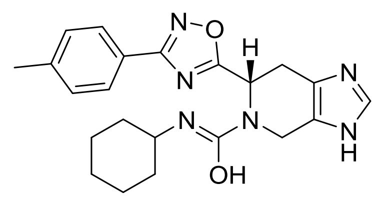
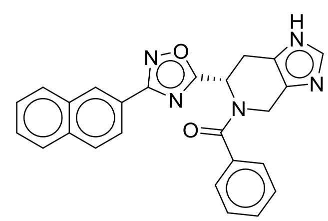
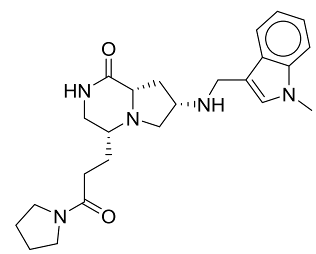
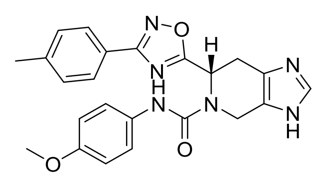
Figure 11. Structural superimposition of the initial protein and four complex structures (before MD simulation) and final complex structures (After MD simulation) obtained from 100 ns MD simulation. (a) MurG protein, (b) MurG- ZINC09186673, (c) MurG- ZINC08439415, (d) MurG-ZINC08856120 and (e) MurG- ZINC08857096 (Color codes: (a) Blue-before MD free MurG model, magenta-after MD free MurG model, (b) Blue-before MD MurG-ZINC09186673 complex, Green-after MD MurG-ZINC09186673 complex, (c) Blue-before MD MurG-ZINC08439415 complex, Red-after MD MurG-ZINC08439415 complex, (d) Blue: before MD MurG-ZINC08856120 complex, Cyan: after MD MurG-ZINC08856120 complex and (e) Blue: before MD MurG-ZINC08857096 complex, slightly brown: MurG-ZINC08857096 complex).

Table 1. Evaluation of MurG (before and after energy minimization) and the template (1F0K).

Proteins Structure	Evaluation Tools								
	Procheck							ERRAT quality factor (%)	Verify_3D (%)
	Ramachandran Plot statistics				G-Factors				
	Most favored (%)	Additional allowed (%)	Generously Allowed (%)	Disallowed (%)	Dihedral angles	Covalent geometry	Overall average		
MurG	90.90	8.20	0.60	0.30	0.08	-0.22	-0.04	66.85	80.55
Minimized MurG by									
YASARA	93.70	5.40	0.60	0.30	0.00	0.39	0.17	99.44	83.84
Chiron	89.90	8.80	0.90	0.30	-0.68	-0.03	-0.42	86.08	82.19
Modrefiner	92.10	6.60	0.60	0.60	0.11	-1.33	-0.49	84.23	86.58
3Drefine	91.50	7.30	0.60	0.60	0.06	-1.46	-0.67	85.99	78.36
Template (1F0K)	94.60	5.40	0.00	0.00	0.29	0.55	0.40	95.02	98.01

Table 2. Binding affinity and inhibition constant (Ki) of twelve (12) selected ligands with their schematic representations.

S/No.	Zinc Code	Estimated Binding energy (kcal/mol)	Estimated Inhibition constant (nM)	Molecular formula	Chemical scheme
1	ZINC08857096	−9.6	91.83	$C_{28}H_{27}NO_5S$	
2	ZINC15707240	−9.6	91.83	$C_{24}H_{20}N_5O_4^+$	
3	ZINC09186673	−9.5	108.69	$C_{26}H_{18}O_6$	
4	ZINC08856120	−9.4	128.70	$C_{28}H_{27}NO_5S$	
5	ZINC08439415	−9.1	213.60	$C_{26}H_{29}NO_5$	

6	ZINC22920899	−9.1	213.60	$C_{28}H_{34}N_6O_2$	
7	ZINC12530134	−9	252.79	$C_{28}H_{23}N_5O_2$	
8	ZINC39960784	−8.9	299.29	$C_{26}H_{35}N_5O$	
9	ZINC15675806	−8.8	354.29	$C_{22}H_{26}N_6O_2$	
10	ZINC15675922	−8.8	354.29	$C_{25}H_{19}N_5O_2$	
11	ZINC22936720	−8.8	354.29	$C_{24}H_{33}N_5O_2$	
12	ZINC15675812	−8.7	419.50	$C_{23}H_{22}N_6O_3$	

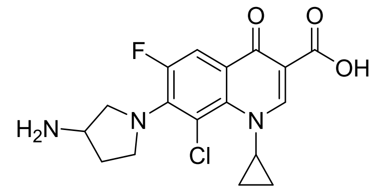
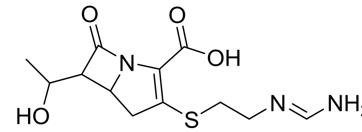
13	CID: 60063	−6.9	8753	$C_{17}H_{17}ClFN_3O_3$	
14	CID: 104838	−5.6	78540	$C_{12}H_{17}N_3O_4S$	

Table 3. Lipinski's screening for twelve (12) selected natural compounds obtained from DataWarrior and PreADMET

SNo	Ligands ID	DataWarrior (Lipinski's rule of five)						PreADMET (Lipinski's Rule of Five)
		MW (≤ 500)	cLogP (≤ 5)	rBonds (≤ 10)	HBA (≤ 10)	HBD (≤ 5)	DL	
1.	ZINC08857096	489.59	4.9263	5	6	0	1.5322	Suitable
2	ZINC15707240	441.446	3.0169	4	9	1	4.7461	Suitable
3	ZINC09186673	424.407	1.1629	3	6	0	-0.0386	Suitable
4	ZINC08856120	489.59	4.9263	5	6	0	1.5322	Suitable
5	ZINC08439415	435.518	3.7452	4	6	0	-1.3382	Suitable
6	ZINC22920899	488.634	-1.9318	5	8	3	6.1471	Suitable
7	ZINC12530134	462.532	2.5076	4	7	2	4.7045	Suitable
8	ZINC39960784	434.606	2.0333	5	6	1	4.2443	Suitable
9	ZINC15675806	406.488	2.6671	3	8	2	1.0168	Suitable
10	ZINC15675922	422.467	1.6989	3	7	2	4.7368	Suitable
11	ZINC22936720	424.567	0.089	6	7	2	6.9105	Suitable
12	ZINC15675812	430.467	2.7501	4	9	2	5.4556	Suitable

Molecular weight (MW), cLogP (Lipophilicity), rBonds (Rotatable bonds), Hydrogen bond donor (HBD), Hydrogen bond acceptor (HBA) and Drug-likeness (DL)

Table 4. ADME parameters of the twelve (12) selected molecules

S/No.	Zinc Code	HIA	BBB	ROCT	cLogS	PSA (Å^2)	Relative PSA(Å^2)	Aqueous Solubility	CYP450 2D6 Inhibitor
1.	ZINC08857096	+	+	NI	-6.437	109.94	0.25418	−4.029	NI
2	ZINC15707240	+	+	NI	-5.556	103.69	0.28087	−2.788	NI
3	ZINC09186673	+	+	NI	-4.917	98.72	0.24317	−3.522	NI
4	ZINC08856120	+	+	NI	-6.437	109.94	0.25418	−4.029	NI
5	ZINC08439415	+	+	NI	-5.616	82.03	0.21657	−4.271	NI
6	ZINC22920899	+	+	NI	-2.413	74.91	0.21122	−3.059	NI
7	ZINC12530134	+	+	NI	-6.221	87.91	0.21879	−2.874	NI
8	ZINC39960784	+	+	NI	-3.030	67.14	0.18323	−2.439	NI
9	ZINC15675806	+	+	NI	-4.718	101.19	0.25735	−3.408	NI
10	ZINC15675922	+	+	NI	-5.397	87.91	0.2446	−3.029	NI
11	ZINC22936720	+	+	NI	-2.297	70.81	0.22853	−2.512	NI
12	ZINC15675812	+	+	NI	-4.669	110.42	0.27731	−3.197	NI

Blood-Brain Barrier (BBB), Human Intestinal Absorption (HIA), Renal Organic Cation Transporter (ROCT), Cytochrome P450 (CYP450 2D6), Water Solubility (cLogS), Polar Surface Area (PSA), Relative Polar Surface Area (Relative PSA), Non-Inhibitor (NI) and Aqueous Solubility = Insoluble < −10 < poorly Soluble < −6 < moderately Soluble < −4 < Soluble −2 < very Soluble < 0 < high Soluble.

Table 5. Toxicity parameters of the twelve (12) selected molecules

S/No.	Zinc Code	AMES (Test) Toxic	Carcinogenic	Mutagenic	Tumorigenic	Reproducibility	Irritant
1	ZINC08857096	No	NC	NM	NT	NR	NI
2	ZINC15707240	No	NC	NM	NT	NR	NI
3	ZINC09186673	No	NC	NM	NT	NR	NI
4	ZINC08856120	No	NC	NM	NT	NR	NI
5	ZINC08439415	No	NC	NM	NT	NR	NI
6	ZINC22920899	No	NC	NM	NT	NR	NI
7	ZINC12530134	No	NC	NM	NT	NR	NI
8	ZINC39960784	No	NC	NM	NT	NR	NI
9	ZINC15675806	No	NC	NM	NT	NR	NI
10	ZINC15675922	No	NC	NM	NT	NR	NI
11	ZINC22936720	No	NC	NM	NT	NR	NI
12	ZINC15675812	No	NC	NM	NT	NR	NI

NC = None Carcinogenic, NM = None Mutagenic, NT = None Tumorigenic, NR = None Reproducibility and NI = None Irritant

Table 6. List of twelve selected ligands with active site residues of MurG through hydrogen bond, pi, hydrophobic and van der Waals interaction.

S/No	Molecule ID	Sources	HBSR	D (Å)	Pi-ISR	D (Å)	HphbISR	vdWISR
1	ZINC08857096	ZINC	Leu275	2.37	Val198	5.12	Asn131,Gly21	Gly200, Arg271,
			Thr276	2.47	Arg170	5.10	Gly20, Phe24	Leu292,Gln300
			Gly273	2.85	Phe24	3.76	Glu279, Arg271	Thr19, His22
			Thr276	2.99	Leu275	4.43	Gly273 Ala274	Asn131, His299
			Gly20	2.80	Leu275	2.62	His299	Ala274, Gly21
			Gly21	2.92	-	-		Gly20, Glu279
2	ZINC15707240	ZINC	His299	2.13	Arg271	5.41	Gln300,Thr276	His22, Asn131
			Gly21 -	3.77	Leu275	5.02	Gly200, Glu279	Gly273, Thr19
			- -	- -	Leu275	4.62	Phe24, Ala274	Thr276, Gly200
				-	Leu275	4.50	Leu275, Gly21	Glu279, Phe24
					Ala274	5.25	Gly273, His22	Gly20, Gln300
							Asn131	Ile294
3	ZINC09186673	ZINC	Thr276	2.53	Leu275	5.11	Glu279, Phe24	Phe24, thr19 Gly21
			Thr276	2.44	Leu275	3.96	Leu275, Gly20	Asn131 Gly273
			Ala274	2.78	Leu275	3.82	Asn131, Ala274	Gln300 Glu279
			Leu275	2.68	Arg271	5.35	Gly21 Gly273	
			Gly20 -	3.48	Arg271	3.61	Arg271	
				-	Cys270	4.73		
4	ZINC088561	ZINC	Thr27	2.4	Val19	5.0	Gly200,	Arg170,

	20		6	8	8	5	Gly21	Gly200
			Gln30	3.0	Leu27	2.6	Arg271,	Arg271,
			0	3	5	4	Gln300	Ala274
			Leu27	2.3	Leu27	4.5	Gly273,	Gly273
			5	2	5	1	His299	His299
			Thr19	2.5	Phe24	3.6	Asn131,	Asn131
			Gly20	9	His22	4	Ala274	Gly21
				2.7		4.6	His22, Thr19	
				0		4	Gly20, Phe24	
5	ZINC084394	ZINC	Thr27	2.4	Leu27	2.6	Leu275,	Gly273
	15		6	5	5	4	Glu279	
			Thr27	3.0	Leu27	4.4	Phe24, Gly20	Gln300
			6	6	5	1		
			Leu27	2.4	Phe24	3.7	Val198,	Asn131
			5	3	Arg17	6	Gly273	Thr19
			Gly21	2.9	0	5.0	Gly200,	Ala274
			Gly20	6	Val19	3	Ala274	Glu279
			-	2.7	8	4.8	Cys270,	Gly199
			-	9	Cys27	4	Gln300	Gly200
			-	-	0	4.1	Gly199	
				-	Arg27	6	Arg271	
				-	1	4.5	Gly21	
					Arg27	8		
					1	3.8		
						8		
6	ZINC229208	ZINC	Nil	Nil	Ala27	4.6	Asn131,	Cys270,
	99				4	5	Gly21	Thr276
			-	-	Leu27	5.3	Ala274,	Met258,
			-	-	5	8	Gln300	Arg170
			-	-	Leu27	5.2	Gly200,	Ser201,
					5	5	Leu275	Gly200
					Phe24	4.9	Arg170,	Gln300,
					Glu27	1	Phe24	Leu292
					9	4.0	Glu279,	Ile294,
					Glu27	1	Thr276	Asn131
					9	3.7	Arg271,	Gly21
						0	Leu292	Ala272
							Gly273	
7	ZINC125301	ZINC	Thr27	2.6	Leu27	5.2	Phe24,	Val169,
	34		6	8	5	2	Glu279	Ser201

			Arg271 - -	2.90 - -	Leu275 Phe24 Phe24 Glu279 Leu275	3.60 4.20 4.62 3.76 5.22	Leu275, His299 Asn131, Gln300 Arg271, Gly200	Gly200, Gly199 Val198, Cys270 Ala272, Gly273 Thr19, Gly21 Asn131, His299 Ile294, Gln300 Leu292, Arg170
8	ZINC39960784	ZINC	Asn131 Thr276 - - - -	1.93 3.56 - - - -	Val198 Met258 Arg271 Ala274 Leu275 Phe24	4.22 5.30 4.44 5.40 4.83 4.89	Val198, Tyr262 His22, Glu279 Leu275, Phe24 Thr276, Arg271 Gly273, Gly21 Ala274, Thr19	Gln130, Gln300 Gly21, His22 Thr19 Gly273 Gly200 Tyr262
9	ZINC15675806	ZINC	Leu275 Ala274 - - - - -	2.43 2.18 - - - - -	His299 His22 His22 Ala274 Arg170 Phe24 Leu275	4.55 5.38 5.10 4.66 4.97 4.53 4.76	Asn131, His22 His299, Gly273 Phe24 Arg170 Glu279 Gly21 Ala274	Glu279 Gln300 Asn131 Gly273 Thr19 Gly21 -
10	ZINC156759	ZINC	Thr27	2.0	Ala27	5.3	Gly20, His22	Phe24,

	22		6 -	7 -	4 Leu275 Leu275 Glu279	4 4.61 5.07 4.07	Asn131, His299 Gln300, Gly21 Ala274, Gly273 Glu279, Leu2775 Phe24	Arg170 Gly20, His22 Asn131, Thr19 Gly21, Gly273 Gln300, His299 Ile294
11	ZINC22936720	ZINC	Thr276 Arg271 - -	2.10 1.96 - -	Leu275 Leu275 Ala274 Phe24	5.15 5.16 4.42 4.37	Val198, Phe24 Glu279, Leu275 Gly21, Ala274 Asn131, Gly273 Cys270	Gln130, Asn131 Gly21, Gln300 Gly273, Glu279 Arg170, Gly200 Val198, Cys270
12	ZINC15675812	ZINC	Thr276 His299 - - -	2.47 2.49 - - -	Arg271 Leu275 Arg170 Phe24 Phe24	4.44 4.30 4.57 5.01 3.67	Leu275, Phe24 Glu279, Asn131 Gln300 Gly273 Arg271	Ile294, Ala274 Gln300, Thr19 Gly21, Gly273 Gly20, Gly200 Glu279, Asn131 His22
13	CID: 60063	PubChem	Asn131 Leu275	2.21 2.06	Leu292	5.24	Gln130, Ile294 Gln300, Thr276 Gly200, Arg271 Gly273, Gly21 Ala274	Ile294, Gln300 Ser201, Leu202 Gly200, Gly199 Arg271, Thr276 Gly273, Ala274

							Gln130, Gly21 Thr19
14	CID: 104838	PubChem	Glu279 Thr276 Thr276 Thr276 Leu275 Gly273 Gln300	1.98 2.38 2.30 2.21 2.74 2.30 2.39	Nil	Leu275 Arg271 Gly200	Ala274 Arg271 Tyr262 Val198 Gly200

HBSR = Hydrogen Bond Sharing Residues, D = Distance, PI-ISR = Pi-Interaction Sharing Residues, HphbISR = Hydrophobic Interaction Sharing Residues and vdWISR = van der Waals Interaction Sharing Residues